Prescribed-Time Robust Synchronization of Networked Heterogeneous Euler-Lagrange Systems

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***Abstract*— In this paper, we propose a prescribed-time synchronization (PTS) algorithm for networked Euler-Lagrange systems subjected to external disturbances. Notably, the system matrix and the state of the leader agent are not accessible to all agents. The algorithm consists of distributed prescribed-time observers and local prescribed-time tracking controllers, dividing the PTS problem into prescribed-time convergence of distributed estimation errors and local tracking errors. Unlike most existing prescribed-time control methods, which achieve prescribed-time convergence by introducing specific time-varying gains and adjusting feedback values, we establish a class of $\mathcal{K}_T$ functions and incorporate them into comparison functions to represent time-varying gains. By analyzing the properties of class $\mathcal{K}_T$ and comparison functions, we ensure the prescribed-time convergence of distributed estimation errors and local tracking errors, as well as the uniform boundedness of internal signals in the closed-loop systems. External disturbances are handled and dominated by the time-varying gains that tend to infinity as time approaches the prescribed time, while the control signal is still guaranteed to be bounded. Finally, a numerical example and a practical experiment demonstrate the effectiveness and innovation of the algorithm.**

***Note to Practitioners*—This paper aims to address the issue of prescribed-time synchronization for networked Euler-Lagrange systems. Existing research on asymptotic and finite-time convergence reveals that the settling time for synchronization is significantly influenced by the system's initial values and controller parameters, making it challenging to be freely pre-designed. In contrast, our proposed prescribed-time synchronization algorithm ensures that all Euler-Lagrange systems achieve synchronization within a prescribed time. The effectiveness of our algorithm has been validated through numerical simulations and physical experiments. In practical applications, our algorithm can be utilized for cooperative control in robotic manipulators and drones. Compared to traditional PD controllers, our proposed algorithm not only offers the advantage of arbitrary settling time configuration in cooperation but also ensures faster response speeds and higher control accuracy, owing to the incorporation of time-varying gains.**

***Index Terms*— Prescribed-time synchronization, Euler-Lagrange systems, comparison functions, multi-agent systems.**

Received 14 October 2024; revised 3 January 2025; accepted 26 January 2025. Date of publication 11 February 2025; date of current version 16 April 2025. This article was recommended for publication by Associate Editor H. Yan and Editor G. Fortino upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 62173155 and Grant 52188102; in part by the Program for the Huazhong University of Science and Technology (HUST) Academic Frontier Youth Team; and in part by the Taihu Lake Innovation Fund for Future Technology, HUST. *(Corresponding author: Lijun Zhu.)*

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Digital Object Identifier 10.1109/TASE.2025.3541052

I. Introduction

In recent years, synchronization (or consensus) of networked Euler-Lagrange systems has received increasing attention due to its great application potential in drone swarms [1] and multi-robot systems [2], etc. Early studies [3], [4], [5], [6], [7], [8] explored the asymptotic synchronization of these systems, achieving synchronization errors approaching zero as time extends infinitely.

Compared with the asymptotic stability theory, finite-time control [9], [10] offers improved convergence performance and enhanced disturbance rejection capabilities. Research on finite-time synchronization of networked Euler-Lagrange systems, as illustrated in [11] and [12], indicates that the settling time has an upper bound that generally depends on initial values and control parameters. Furthermore, studies on fixed-time synchronization [13], [14] indicate that the finite settling time is invariant with respect to initial values, although control parameters still play a role. Practical fixed-time synchronization, introduced in [15], ensures convergence within a fixed time to a neighborhood of zero synchronization error. According to [16] and [17], both finite-time and fixed-time synchronization for multi-agent systems (MASs) are impacted by external disturbances, making the upper bound of the settling time uncomputable in advance. Alternatively, if these disturbances do not alter the settling time, only practical fixed-time convergence is typically achieved, resulting in convergence to a neighborhood of zero rather than exact zero-error, as noted in [15] and [18]. Consequently, neither the upper bound nor the precise settling time can be estimated prior to system operation. Moreover, without disturbances, determining settling time requires awareness of each agent's

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initial values [19]. Thus, these approaches may fall short when specific settling time constraints are required, primarily due to unforeseen disturbances and locality of agents' initial values [20].

As an extension of finite-time control, prescribed-time control is introduced in [21], where the settling time is prescribed as the name suggests and independent of initial values and any other design parameters. Its application to a single Euler-Lagrange system is discussed in [22], and it has been further extended to various MASs. Examples include linear MASs [23], single-integrator MASs [24], [25], double-integrator MASs [26], chain-integrator MASs [27], and strict-feedback MASs [28], [29]. Prescribed-time control has also been applied to different forms of synchronization. These include synchronization in the leaderless case [30], the leader-follower case [31], containment control [32], and formation control [33], among others. Notably, practical prescribed-time consensus for chain-integrator MASs is proposed in [29], in which fuzzy logic and the adaptive method are adopted to handle dead-zone input and sensor faults. Moreover, an event-triggered communication mechanism-based prescribed-time consensus is proposed in [34], showing that practical prescribed-time convergence still holds under discrete-time communication.

Although prescribed-time control has been successfully extended to various forms of MASs and synchronization control types, the specific solution for prescribed-time synchronization in networked Euler-Lagrange systems within a leader-follower framework remains unexplored. This gap in the literature motivates our study. Current studies on prescribed-time control for MASs, as seen in [23], [24], [25], [26], [27], [28], and [29], predominantly focus on systems without external disturbances, neglecting the aspects of robustness and disturbance rejection through prescribed-time control. While some studies, like [29] and [34], propose practical prescribed-time control with time-varying gains that have finite upper bounds, these only ensure convergence to a neighborhood of zero. Our approach diverges from traditional finite-time [11], [12] and fixed-time synchronization methods [13], [14], which rely on fractional-power feedback. Instead, we accomplish prescribed-time convergence of synchronization errors by implementing time-varying gains that grow indefinitely. This ensures that the settling time is independent of initial values and design parameters, providing uniform convergence across all agents. However, given the complexity inherent to high-order nonlinear systems in prescribed-time control, and the potential destabilization from unbounded time-varying gains, particularly in the presence of external disturbances, this task is challenging.

In this paper, we address the robust PTS problem for networked Euler-Lagrange systems subjected to external disturbances. The main features and contributions are summarized as follows: (i) We develop distributed prescribed-time observers for followers to estimate the leader system's information, including its state and system matrix. This allows us to tackle the PTS problem by addressing prescribed-time stabilization of both distributed estimation errors and local tracking errors. Distributed controllers are then designed using time-varying state transformations and sliding mode control. (ii) Contrary to existing prescribed-time control methods that rely on introducing specific time-varying gains, we construct a class of $\mathcal{K}_T$ functions and integrate them into a comparison function. These are incorporated into the feedback loop as time-varying gains. This approach is theoretically more general, offering greater design flexibility and enabling various forms of prescribed-time convergence. (iii) In the presence of external disturbances, unbounded time-varying gains may excessively amplify these disturbances, potentially causing unbounded internal signals in closed-loop systems. To address this, we propose sufficient conditions for combining the class $\mathcal{K}_T$ and comparison functions. These conditions ensure a sufficient prescribed-time convergence rate, achieving not only PTS but also guaranteeing the boundedness of all internal signals in closed-loop MASs, thus mitigating the impact of external disturbances on prescribed-time convergence.

The rest of the paper is organized as follows. Section II gives the notations and problem formulation. Section III constructs the $\mathcal{K}_T$ and $\mathcal{D}_T$ functions. The main results of the proposed PTS algorithm are presented in Section IV. Numerical and experimental results are performed in Section V. Finally, Section VI concludes this paper.

II. Notations and Problem Formulation

A. Notations

$\mathbb{R}$, $\mathbb{R}_+$, and $\mathbb{R}^n$ denote the set of real numbers, the set of non-negative real numbers, and the n-dimensional Euclidean space, respectively. $\mathbb{R}^{n\times m}$ denotes the set of real matrices with dimensions n by m. t_0 denotes the initial time, $T > 0$ denotes the prescribed time, and $\mathcal{T}_p$ denotes the time interval $\mathcal{T}_p = \{t : t_0 \leq t < T + t_0\}$. For a positive definite matrix A, $\underline{\lambda}_A$ and $\bar{\lambda}_A$ denote the minimum and maximum eigenvalues of A, respectively. $|A|_2 = \max\{\lambda(A^\top A)\}$ denotes the 2-norm of a matrix A with $\lambda(A^\top A)$ being the spectrum of $A^\top A$. For a matrix $s = [s_1, \cdots, s_n] \in \mathbb{R}^{n\times n}$ where $s_i \in \mathbb{R}^n$ for $i = 1, \cdots, n$, $\mathrm{vec}(s) = [s_1^\top, \cdots, s_n^\top]^\top \in \mathbb{R}^{nn}$. For a $\mathcal{C}^k$ function (i.e., k-times continuously differentiable) $\mu(t) : \mathbb{R} \to \mathbb{R}$, $\mathrm{d}^k\mu(t)/\mathrm{d}t^k$ is its k-th derivative with respect to t.

B. Multi-Agent Systems

Consider a group of n-link robotic manipulators described by the Euler-Lagrange dynamics, for $i = 1, \cdots, N$,

$$\begin{aligned} \dot{q}_i &= r_i, \\ \dot{r}_i &= M_i^{-1}(q_i)(\tau_i - d_i(t) - C_i(q_i, r_i)r_i - G_i(q_i)) \end{aligned} \tag{1}$$

where $q_i \in \mathbb{R}^n$ and $r_i \in \mathbb{R}^n$ are the vectors of generalized position and velocity of the i-th robotic manipulator, respectively. The symmetric and positive definite matrix $M_i(q_i) : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is the inertia matrix, $C_i(q_i, r_i) : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is the matrix related to the Coriolis and centrifugal forces, $G_i(q_i) : \mathbb{R}^n \to \mathbb{R}^n$ is the gravitational torque, $d_i(t) \in \mathbb{R}^n$ denotes the external disturbance, and $\tau_i \in \mathbb{R}^n$ is the vector of control forces. According to [35], some properties and assumptions for system (1) are presented as follows.

Property 1: $M_i(q_i)$, $C_i(q_i, r_i)$ and $G_i(q_i)$ in (1) satisfy the following properties: $k_{\underline{n}} I_n \leq M_i(q_i) \leq k_{\bar{n}} I_n$, $\|C_i(q_i, r_i)\| \leq$

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$k_c\|r_i\|$, and $\|G_i(q_i)\| \le k_g$ for $q_i, r_i \in \mathbb{R}^n$ and $i = 1, \cdots, N$, where $k_{\underline{n}}$, $k_{\bar{n}}$, k_c and k_g are some positive constants. ∎

Assumption 1: $d_i(t)$ satisfies $\sup_{t\in[t_0,\infty)} \|d_i(t)\| \le \bar{d}_i$ for $i = 1, \cdots, N$, where $\bar{d}_i$ is a positive, finite constant that does not need to be known. ∎

We call the i-th robotic manipulator agent i. The reference trajectory is generated by a system, labeled agent 0, and described by

$$\dot{\upsilon}_0 = S_0\upsilon_0 \tag{2}$$

where $S_0 \in \mathbb{R}^{n\times n}$ is a constant matrix, and $\upsilon_0 \in \mathbb{R}^n$ is the desired trajectory to be tracked.

We first review some terminologies in graph theory to describe the interaction among agents. Denote a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with the node set $\mathcal{V} = \{0, 1, \cdots, N\}$ and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The existence of the edge $(i, j) \in \mathcal{E}$ indicates that agent j can obtain information from the i-th agent. Denote the node set $\bar{\mathcal{V}} = \{1, \cdots, N\}$ without node 0. Agent j is called a neighbor of i if $(j, i) \in \mathcal{E}$. The set of neighbors of agent i is denoted by $\mathcal{N}_i = \{j \in \mathcal{V} : (j, i) \in \mathcal{E}\}$. Denote by $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{(N+1)\times(N+1)}$ the weighted adjacency matrix of $\mathcal{G}$, where $a_{ij} > 0$ if $(j, i) \in \mathcal{E}$ and $a_{ij} = 0$, otherwise. Self-edges are not allowed, i.e., $a_{ii} = 0$. The Laplacian matrix of the graph is denoted as $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{(N+1)\times(N+1)}$, where $l_{ii} = \sum_{j=0}^{N} a_{ij}$, and $l_{ij} = -a_{ij}$ with $i \ne j$.

For the MASs consisting of (1) and (2), we associate node 0 with system (2) and call it the leader. The other nodes are associated with system (1) and are called followers. In this paper, we assume S_0 and υ_0 are known only to the neighbors of the leader system.

Assumption 2: The graph $\mathcal{G}$ contains a spanning tree with node 0 as the root. ∎

Remark 1: Let $\Delta = \mathrm{diag}\{a_{10}, \cdots, a_{N0}\}$, then the Laplacian $\mathcal{L}$ of $\mathcal{G}$ can be expressed as

$$\mathcal{L} = \left[\begin{array}{c|c} \sum_{j=1}^{N} a_{0j} & -(a_{01}, \cdots, a_{0N}) \\ \hline -\Delta 1_N & H \end{array}\right].$$

As shown in [36], under Assumption 1, all elements of $\lambda(H)$ have positive real parts, and H satisfies $H1_N = \Delta 1_N$. ∎

C. Problem Formulation

To the best of our knowledge, the prescribed-time control for nonlinear systems is generally based on three approaches. The time-varying state transformation method [21] converts the prescribed-time convergence of original states into a boundedness problem of the transformed variables. The time-varying gain method [37] increases the control gain to infinity as time approaches the prescribed time instance. The last approach is essentially a finite-time control method, which achieves prescribed-time stability through a fractional-power state feedback and a suitable selection of control parameters [38]. This method has been successfully applied to first-order systems and some canonical systems, while its effectiveness has not been verified for nonlinear systems with a relative degree greater than one. Moreover, this method can only guarantee that the upper bound of settling time is prescribed, while the actual settling time is also affected by initial values and control parameters, leading to inconsistency across agents. Therefore, in this paper, we adopt the first two methods where a class of functions will be useful for the time-varying state transformation and the prescribed-time control later, and it is defined as follows.

Definition 1: A twice differentiable function $\mu : \mathcal{T}_p \to [a_0, \infty)$ is said to belong to the class $\mathcal{K}_T$ if it is strictly increasing with

$$\frac{\mathrm{d}^k\mu(t)}{\mathrm{d}t^k}\Big|_{t=t_0} = a_k, \quad \lim_{s\to t_0+T} \frac{\mathrm{d}^k\mu(t)}{\mathrm{d}t^k}\Big|_{t=s} = \infty$$

for $\forall k = 0, 1, 2$, where a_k is a positive constant possibly related to T and there exist finite numbers $w_k > 0$ and $\iota_k > 1$ such that

$$\mu^{(k)}(t) \le w_k\big(\mu^{(k-1)}(t)\big)^{\iota_k}, \quad \forall t \in \mathcal{T}_p,\ k = 1, 2. \tag{3}$$

A function $\varphi : \mathcal{T}_p \to (0, b]$ is said to belong to class $\mathcal{D}_T$, if it is strictly decreasing with $\varphi(t_0) = b$ and $\lim_{t\to t_0+T} \varphi(t) = 0$ where b is a positive constant possibly related to T. ∎

Remark 2: A few examples of $\mathcal{K}_T$ functions are $\mu(t) = (T + t_0 - t)^{-m}$, $\mu(t) = \exp\big((T + t_0 - t)^{-1}\big)$ and $\mu(t) = \zeta^{1/(T+t_0-t)}$ with some constants $\zeta > 1$ and $m > 0$. ∎

A class of functions is defined as an extension to $\mathcal{K}_\infty$ and $\mathcal{KL}$ functions (Definitions 4.2 and 4.3 in [39]) that have different domains.

Definition 2: A function $\alpha : [0, \infty) \to [a, \infty)$ is said to belong to class $\mathcal{K}_\infty^e$ if it is strictly increasing and $\alpha(0) = a > 0$. A continuous function $\beta : [0, c) \times \mathcal{T}_p \to [0, \infty)$ is said to belong to class $\mathcal{KL}_T$ if, for each fixed s, the mapping $\beta(r, s)$ belongs to class $\mathcal{K}$ with respect to r and, for each fixed r, the mapping $\beta(r, s)$ is decreasing with respect to s and $\beta(r, s) \to 0$ as $s \to t_0 + T$. ∎

We then give the definition of prescribed-time convergence as follows.

*Definition 3 (*Prescribed-Time Convergence*):* Consider the system

$$\begin{aligned} \dot{x} &= f(x, d(t), t), \quad x(t_0) = x_0, \\ e &= h(x, d(t), t) \end{aligned}$$

where $x \in \mathbb{R}^n$ is the state, $d(t) : [t_0, \infty) \to \mathbb{R}^{n_d}$ is the external input, $e \in \mathbb{R}^p$ is the output, and x_0 is the initial state at the initial time t_0. For a given $T > 0$, the system output is said to achieve prescribed-time convergence (towards zero within T) if there exists a $\mathcal{KL}_T$ function β such that, for any x_0 and any $d(t)$,

$$\|x(t)\| \le \beta(\|\bar{x}_0\|, t),\ \forall t \in \mathcal{T}_p$$

where $\|\bar{x}_0\| = \sqrt{\|x_0\|^2 + \|d(t)\|^2_{[t_0,\infty)}}$. ∎

This paper aims to propose a distributed control law in the form of

$$\begin{aligned} \dot{S}_i &= \sigma_{S,i}\big(\textstyle\sum_{j\in\mathcal{N}_i} a_{ij}(S_j - S_i), \mu(t)\big), \\ \dot{\upsilon}_i &= \sigma_{\upsilon,i}\big(\textstyle\sum_{j\in\mathcal{N}_i} a_{ij}(\upsilon_j - \upsilon_i), S_i, \mu(t)\big), \\ \tau_i &= \sigma_{\tau,i}(q_i, r_i, \upsilon_i, \dot{\upsilon}_i, \mu(t)),\ i \in \bar{\mathcal{V}} \end{aligned} \tag{4}$$

with functions $\sigma_{S,i}$, $\sigma_{\upsilon,i}$, and $\sigma_{\tau,i}$ to be designed and $\mu(t) \in \mathcal{K}_T$. Since the state and matrix of the leader (2) are only

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accessible to a subset of followers via the communication network, S_i- and υ_i-dynamics in (4) are constructed for each follower to obtain the estimation of the leader's state υ_0 and matrix S_0, respectively. Let $e_i = q_i - \upsilon_0$ and $e = [e_1^\top, \cdots, e_N^\top]^\top$. Compared with the exponential synchronization in [36] and fixed-time synchronization in [14], we introduce the time-varying transformation for the prescribed-time stability and thus $\mu(t)$ is introduced in (4). Then, the prescribed-time synchronization problem is defined as follows.

Prescribed-Time Synchronization Problem: Given the system composed of (1) and (2) that communicate through directed graph $\mathcal{G}$. For a prescribed time $T > 0$, design a distributed control law of the form (4) such that, for all bounded initial values, the synchronization error e of the closed-loop system achieves prescribed-time convergence in the sense of Definition 3 and all internal signals are bounded.

III. $\mathcal{K}_T$ AND $\mathcal{D}_T$ FUNCTIONS

In this section, we discuss the properties of $\mathcal{K}_T$ and $\mathcal{D}_T$ functions and their involvement in the Lyapunov stability analysis. We simplify $\mu(t)$ and $\dot{\mu}(t)$ to μ and $\dot{\mu}$ throughout this paper if no confusion arises. For $\gamma \in \mathcal{K}_\infty$, the incorporation of class $\mathcal{K}_T$ and comparison functions is defined as $\gamma(\mu)$. Define

$$\kappa^\iota(\gamma(\mu)) = \exp\left(\iota \int_{t_0}^{t} \gamma(\mu(s))\mathrm{d}s\right),\ t \geq t_0, \tag{5}$$

$$\kappa^\iota(\gamma(\mu(\zeta))) = \exp\left(\iota \int_{t_0}^{\zeta} \gamma(\mu(s))\mathrm{d}s\right),\ \zeta \geq t_0 \tag{6}$$

where ι is a constant and $\gamma \in \mathcal{K}_\infty$. We then have the following propositions and lemmas. *The proof of Proposition 1 is straightforward and omitted here, while the other proofs are listed the in appendix.*

Proposition 1: Let $\mu \in \mathcal{K}_T$ and $\gamma \in \mathcal{K}_\infty$. Then $\kappa^{-1}(\gamma(\mu))$ is a $\mathcal{D}_T$ function satisfying $\kappa^{-1}(\gamma(\mu)) < 1$. If $\alpha \in \mathcal{K}_\infty$ or $\alpha \in \mathcal{K}_\infty^e$, $\beta(r,t) = \alpha(r)\kappa^{-c}(\gamma(\mu))$ for any $c > 0$ is a $\mathcal{KL}_T$ function. ∎

Proposition 2: Let $\mu \in \mathcal{K}_T$ with some ι_1 and w_1 satisfying (3). For some $\gamma \in \mathcal{K}_\infty$ and any $c_1, c_2 > 0$, the function $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))$ belongs to class $\mathcal{D}_T$ for $t \in \mathcal{T}_p$ if the function $\gamma(s)$ satisfies

$$\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s} \leq \varepsilon w_1^{-1}\frac{c_2}{c_1}s^{-\iota_1}(\gamma(s))^2 \tag{7}$$

for $s \in [a_0, \infty)$ and $0 < \varepsilon < 1$. Moreover, $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu)) \leq \epsilon_1\kappa^{-(1-\varepsilon)c_2}(\gamma(\mu))$ where $\epsilon_1 = (\gamma(a_0))^{c_1}$ and $a_0 = \mu(t_0)$. ∎

Proposition 3: Let $\mu \in \mathcal{K}_T$ with some ι_1, ι_2, w_1 and w_2 satisfying (3). Define

$$\bar{\gamma}(t) = \dot{\gamma}(\mu(t)) + \gamma^2(\mu(t)) + \gamma(\mu(t)). \tag{8}$$

Suppose the conditions of Proposition 2 hold with $\gamma(s)$ further satisfying

$$\frac{\mathrm{d}^2\gamma(s)}{\mathrm{d}s^2} + \frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}w_2 w_1^{\iota_2-1}s^{\iota_1(\iota_2-1)} \leq \varepsilon c\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\gamma(s) \tag{9}$$

for $s \in [a_0, \infty)$ and $0 < \varepsilon < 1$. Then, for any $c \geq 2c_2/c_1$, $\bar{\gamma}(t)\kappa^{-c}(\gamma(\mu))$ belongs to class $\mathcal{D}_T$ for $t \in \mathcal{T}_p$. Moreover, $\bar{\gamma}(t)\kappa^{-c}(\gamma(\mu)) \leq \epsilon_2\kappa^{-\frac{2(1-\varepsilon)c_2}{c_1}}(\gamma(\mu))$ where $\epsilon_2 = \dot{\gamma}(\mu(t_0)) + \gamma(a_0) + (\gamma(a_0))^2$. ∎

Lemma 1: Let $\mu \in \mathcal{K}_T$ and $\gamma_1, \gamma_2 \in \mathcal{K}_\infty$. Consider a system $\dot{x} = f(x,d)$ where $x \in \mathbb{R}^n$ and $d \in \mathbb{R}^{n_d}$. If there exists a positive definite function $V(x) = x^\top Px$ where $P > 0$ such that its derivative along x-dynamics satisfies

$$\dot{V} \leq -(c_1\gamma_1(\mu(t)) - c_2)V + c_3\kappa^{-c_1}(\gamma_2(\mu)) \tag{10}$$

for any positive numbers $c_1 > 0$, $c_2, c_3 \in \mathbb{R}$ and $\gamma_1(s) - \gamma_2(s) < 0$ for $s \in [a_0, \infty)$, then $\|x(t)\| \leq \alpha(\|x(t_0)\|)\kappa^{-c_1/2}(\gamma_1(\mu))$, where α is a $\mathcal{K}_\infty^e$ function if $c_3 \neq 0$, and otherwise a $\mathcal{K}_\infty$ function. ∎

Lemma 2: Let $\mu \in \mathcal{K}_T$ and $\gamma \in \mathcal{K}_\infty$. Consider a system $\dot{x} = f(x,d)$ where $x \in \mathbb{R}^n$ and $d \in \mathbb{R}^{n_d}$. If there exists a positive definite function $V(x) = x^\top Px$ where $P > 0$ such that its derivative along x-dynamics satisfies

$$\dot{V} \leq -c_1\gamma(\mu)V + c_2\gamma(\mu) \tag{11}$$

for some constants $c_1 > 0$, $c_2 \in \mathbb{R}$, then $\|x(t)\| \leq \alpha(\|x(t_0)\|)$, where α is a $\mathcal{K}_\infty^e$ function. ∎

Remark 3: There are many choices for $\gamma(s)$ to satisfy the conditions in Proposition 2 and 3. For example, for a given $\mu \in \mathcal{K}_T$, any $0 < \varepsilon < 1$, and $c_1, c_2 > 0$, one can find $m \geq \max\{\iota_1 - 1, \iota_1(\iota_2 - 1)\}$, $k > c_1 m\omega_1/(a_0^{m+1-\iota_1}c_2\varepsilon)$ and $c > \max\{2\omega_1^{\iota_2-1}\omega_2/(\varepsilon k a_0^{m-\iota_1(\iota_2-1)}), 2(m-1)/(k\varepsilon a_0^{m+1})\}$, where $a_0 = \mu(t_0)$ and ι_i, ω_i are given in Definition 1. Let $\gamma(s) = ks^m$. We can check W $\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s} = \varepsilon\omega_1^{-1}c_2c_1^{-1}s^{-\iota_1}(\gamma(s))^2\varepsilon^{-1}\omega_1c_1c_2^{-1}mk^{-1}s^{\iota_1-m-1} \leq \varepsilon w_1^{-1}\frac{c_2}{c_1}s^{-\iota_1}(\gamma(s))^2$ and $\frac{\mathrm{d}^2\gamma(s)}{\mathrm{d}s^2} + \frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}w_2w_1^{\iota_2-1}s^{\iota_1(\iota_2-1)} = k^2m(m-1)s^{2m-2} + km\omega_2\omega_1^{\iota_2-1}s^{m-1+\iota_1(\iota_2-1)} \leq \varepsilon c\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\gamma(s)$. Thus, the conditions in Propositions 2 and 3 are satisfied. ∎

Remark 4: The motivation behind Propositions 1 and 2 is to analyze the prescribed-time convergence of a nominal system. Consider, for instance, the single-integrator system described by $\dot{x} = u$, where $x \in \mathbb{R}$ is the state and $u \in \mathbb{R}$ is the control input. Suppose the control input is given by $u = -c\gamma(\mu)x$, with $c > 0$ and $\mu \in \mathcal{K}_T$. Proposition 1 asserts that x achieves the prescribed-time convergence to zero for any $c > 0$. To guarantee the boundedness of u, it is necessary to demonstrate that $\gamma(\mu)\kappa^{-c}(\gamma(\mu)) < \infty$. Proposition 2 provides that this condition is met if $\gamma(s)$ satisfies $\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s} < w_1^{-1}cs^{-\iota_1}(\gamma(s))^2$. Additionally, Proposition 3 is employed to establish prescribed-time convergence in more complex scenarios. ∎

Remark 5: In this paper, both Propositions 2 and 3 are integral to proving the prescribed-time convergence of relevant states and the uniform boundedness of terms associated with μ. Lemma 1 sets the conditions on the Lyapunov function necessary for deriving prescribed-time convergence. Meanwhile, Lemma 2 is utilized to validate the boundedness of the new state after implementing time-varying state transformations within the closed-loop system. ∎

IV. MAIN RESULTS

In this section, we first establish distributed prescribed-time observers to estimate the leader's state υ_0 and the system matrix S_0, which will be used to construct prescribed-time synchronization controllers. Second, we design local prescribed-time tracking controllers by utilizing the

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time-varying state transformation and sliding-mode control method, and prove the prescribed-time convergence of the synchronization errors.

A. Prescribed-Time Observer

We construct the distributed prescribed-time observers as follows

$$\dot{S}_i = \gamma_s(\mu(t))\textstyle\sum_{j\in\mathcal{V}} a_{ij}(S_j - S_i), \tag{12}$$

$$\dot{\upsilon}_i = S_i\upsilon_i + \gamma_\upsilon(\mu(t))\textstyle\sum_{j\in\mathcal{V}} a_{ij}(\upsilon_j - \upsilon_i),\ \ i \in \bar{V} \tag{13}$$

where γ_s, γ_υ are differentiable $\mathcal{K}_\infty$ functions to be designed and μ is a $\mathcal{K}_T$ function. We note $\gamma_s(\mu(t))$ and $\gamma_\upsilon(\mu(t))$ are $\mathcal{K}_T$ functions.

In (12) and (13), S_i and υ_i denote the i-th agents' estimations for the leader's system matrix and state, respectively. Let $S = \left[S_1^\top, \cdots, S_N^\top\right]^\top$ and $\upsilon = \left[\upsilon_1^\top, \cdots, \upsilon_N^\top\right]^\top$. Denote $\tilde{S}_i = S_i - S_0$ and $\tilde{\upsilon}_i = \upsilon_i - \upsilon_0$ as the i-th local estimation errors for S_i and υ_i, respectively. Let

$$\tilde{S} = \left[\tilde{S}_1^\top, \cdots, \tilde{S}_N^\top\right]^\top, \quad \tilde{\upsilon} = \left[\tilde{\upsilon}_1^\top, \cdots, \tilde{\upsilon}_N^\top\right]^\top \tag{14}$$

as the extended estimation errors. Taking time derivative of $\tilde{S}$ yields

$$\begin{aligned}\dot{\tilde{S}} &= -\gamma_\upsilon(\mu)(H\otimes I_n)S + \gamma_\upsilon(\mu)(\Delta\otimes I_n)(1_N\otimes S_0)\\ &= -\gamma_\upsilon(\mu)(H\otimes I_n)S + \gamma_\upsilon(\mu)(\Delta 1_N)\otimes(I_n S_0)\\ &= -\gamma_\upsilon(\mu)(H\otimes I_n)S + \gamma_\upsilon(\mu)(H 1_N)\otimes(I_n S_0)\\ &= -\gamma_\upsilon(\mu)(H\otimes I_n)S + \gamma_\upsilon(\mu)(H\otimes I_n)(1_N\otimes S_0)\\ &= -\gamma_s(\mu)(H\otimes I_n)\tilde{S}.\end{aligned} \tag{15}$$

where we used $\Delta 1_N = H 1_N$ (see Remark 1). Similarly, taking time derivative of $\tilde{\upsilon}$ yields $\dot{\tilde{\upsilon}} = -\gamma_\upsilon(\mu)(H\otimes I_n)\tilde{\upsilon} + \Lambda_s\tilde{\upsilon} + \Lambda_{\tilde{s}}(1_N\otimes\upsilon_0)$ where

$$\Lambda_s = \mathrm{diag}\{S_1, \cdots, S_N\},\ \ \Lambda_{\tilde{s}} = \mathrm{diag}\{\tilde{S}_1, \cdots, \tilde{S}_N\}. \tag{16}$$

By Remark 1, there always exist positive matrices P and Q such that $-P(H\otimes I_n) - (H\otimes I_n)^\top P = -Q$ holds. Define $\zeta = \underline{\lambda}_Q\bar{\lambda}_P^{-1}$. Then, we have the following lemma.

Lemma 3: Consider the leader (2) and observers (12)-(13) in a directed graph $\mathcal{G}$

under Assumption 2. Suppose $\mu \in \mathcal{K}_T$, and γ_s, γ_υ satisfy $\gamma_\upsilon(s) - \gamma_s(s) < 0, \forall s \in [a_0, \infty)$ with $a_0 = \mu(t_0)$. Then, $\tilde{S}(t)$ and $\tilde{\upsilon}(t)$ in (14) achieve prescribed-time convergence toward zero within $T + t_0$, i.e., there exist $\mathcal{KL}_T$ functions β_s and β_υ such that

$$\|\tilde{S}(t)\| \le \beta_s(\|\tilde{S}(t_0)\|, t) = \alpha_s(\|\tilde{S}(t_0)\|)\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu)), \tag{17}$$

$$\|\tilde{\upsilon}(t)\| \le \beta_\upsilon(\|\tilde{\upsilon}(t_0)\|, t) = \alpha_\upsilon(\|\tilde{\upsilon}(t_0)\|)\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu)) \tag{18}$$

where $\alpha_s \in \mathcal{K}_\infty$ and $\alpha_\upsilon \in \mathcal{K}_\infty^e$, and $\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu))$ and $\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu))$ are defined in the sense of (5). Moreover, $S(t)$ and $\upsilon(t)$ are bounded for $t \in \mathcal{T}_p$, i.e.,

$$\|S(t)\| \le \bar{s} = \beta_s(\|\tilde{S}(t_0)\|, t_0) + \|1_N\otimes S_0\|, \tag{19}$$

$$\|\upsilon(t)\| \le \bar{\upsilon} = \beta_\upsilon(\|\tilde{\upsilon}(t_0)\|, t_0) + \sup_{t\in\mathcal{T}_P}\|1_N\otimes\upsilon_0(t)\|, \tag{20}$$

where $\bar{s}$ and $\bar{\upsilon}$ are some positive finite constants. ∎

Proof: First, we rewrite (15) as

$$\frac{\mathrm{d}\mathrm{vec}(\tilde{S})}{\mathrm{d}t} = -\gamma_s(\mu(t))(I_n\otimes H\otimes I_n)\mathrm{vec}(\tilde{S}). \tag{21}$$

Define the Lyapunov candidate functions as $V(\tilde{S}) = \mathrm{vec}(\tilde{S})^\top(I_n\otimes P)\mathrm{vec}(\tilde{S})$ and $U(\tilde{\upsilon}) = \tilde{\upsilon}^\top P\tilde{\upsilon}$, where $P \in \mathbb{R}^{Nn\times Nn}$ is a positive definite matrix.

Due to (21), $\dot{V}(\tilde{S}) \le -\zeta\gamma_s(\mu(t))V(\tilde{S})$. Invoking the comparison lemma yields $V(\tilde{S}(t)) \le \kappa^{-\zeta}(\gamma_s(\mu))V(\tilde{S}(t_0))$, which implies $\|\mathrm{vec}(\tilde{S}(t))\| \le \bar{\lambda}_P\underline{\lambda}_P^{-1}\|\mathrm{vec}(\tilde{S}(t_0))\|\kappa^{-\zeta/2}(\gamma_s(\mu))$. Note that $\|\tilde{S}\| = \theta\|\tilde{S}\|_F = \theta\|\mathrm{vec}(\tilde{S})\|$ for some $0 < \vartheta < \infty$, where $\|\tilde{S}\|_F$ denotes the Frobenius norm of matrix $\tilde{S}$. Therefore, (17) holds with $\alpha_s(\|\tilde{S}(t_0)\|) = \bar{\lambda}_p\underline{\lambda}_P^{-1}\|\tilde{S}(t_0)\|$.

Thus, the prescribed-time convergence is achieved for $\tilde{S}$. Noting that $\beta_s(\|\tilde{S}(t_0)\|, t)$ is strictly decreasing with respect to t, then $\|\tilde{S}(t)\|$ has an upper bound $\beta_s(\|\tilde{S}(t_0)\|, t_0)$ and tends to zero as $t \to T + t_0$, that is $\lim_{t\to T+t_0} S_i(t) = S_0, i \in \bar{V}$. Due to $\tilde{S}(t) = S(t) - 1_N\otimes S_0$, one has $\|S(t)\| \le \|\tilde{S}(t)\| + \|1_N\otimes S_0\|$. Therefore, $S_i(t)$ is bounded for $t \in \mathcal{T}_p$ as in (19). The time derivative of $U(\tilde{\upsilon})$ satisfies

$$\begin{aligned}\dot{U}(\tilde{\upsilon}) &= -\gamma_\upsilon(\mu(t))\tilde{\upsilon}^\top Q\tilde{\upsilon} + 2\tilde{\upsilon}^\top P(\Lambda_s\tilde{\upsilon} + \Lambda_{\tilde{s}}\bar{\upsilon}_0)\\ &\le -\gamma_\upsilon(\mu(t))\underline{\lambda}_Q\|\tilde{\upsilon}\|^2 + 2\tilde{\upsilon}^\top P(\Lambda_s\tilde{\upsilon} + \Lambda_{\tilde{s}}\bar{\upsilon}_0)\end{aligned}$$

where $\bar{\upsilon}_0 = 1_N\otimes\upsilon_0(t)$.

Due to (17) and (19), one has $2\tilde{\upsilon}^\top P\Lambda_s\tilde{\upsilon} \le 2\bar{\Lambda}_s\|P\|\|\tilde{\upsilon}\|^2$ with $\bar{\Lambda}_s = \sup_{t\in\mathcal{T}_p}\|\Lambda_s(t)\|$, where we note $\bar{\Lambda}_s < \infty$ due to (19). Furthermore, $2\tilde{\upsilon}^\top P\Lambda_{\tilde{s}}(1_N\otimes\upsilon_0) \le \|\tilde{\upsilon}\|^2 + \|P\|^2\left(\sup_{t\in\mathcal{T}_p}\|\bar{\upsilon}_0(t)\|\right)^2\alpha_s^2(\|\tilde{S}(t_0)\|)\kappa^{-\zeta}(\gamma_s(\mu))$. Therefore, $\dot{U}$ satisfies $\dot{U}(\tilde{\upsilon}) \le -(\zeta\gamma_\upsilon(\mu(t)) - c)U(\tilde{\upsilon}) + d\kappa^{-\zeta}(\gamma_s(\mu))$, where $c = \left(2\bar{\Lambda}_s\|P\| + 1\right)\underline{\lambda}_P^{-1}$ and $d = \|P\|^2\left(\sup_{t\in\mathcal{T}_p}\|\bar{\upsilon}_0(t)\|\right)^2\alpha_s^2(\|\tilde{S}(t_0)\|)$.

Applying Lemma 1 with $c_1 = \zeta$, $c_2 = c$, $c_3 = d$, $\gamma_1(\mu) = \gamma_\upsilon(\mu)$ and $\kappa^{-1}(\gamma_2(\mu)) = \kappa^{-1}(\gamma_s(\mu))$ leads to $\|\tilde{\upsilon}(t)\| \le \alpha_\upsilon(\tilde{\upsilon}(t_0))\kappa^{-\zeta/2}(\gamma_\upsilon(\mu))$, where $\alpha_\upsilon \in \mathcal{K}_\infty^e$ can be calculated according to the proof of Lemma 1. Therefore, (18) as well as the prescribed-time convergence is achieved for $\tilde{\upsilon}$. Since $\beta_\upsilon(\|\tilde{\upsilon}(t_0)\|, t)$ is strictly decreasing for $t \in \mathcal{T}_p$, then $\tilde{\upsilon}(t)$ has an upper bound $\beta_\upsilon(\|\tilde{\upsilon}(t_0)\|, t_0)$. Due to $\tilde{\upsilon}(t) = \upsilon(t) - 1_N\otimes\upsilon_0(t)$, one obtains $\|\upsilon(t)\| \le \|\tilde{\upsilon}(t)\| + \|1_N\otimes\upsilon_0(t)\|$. Thus $\upsilon(t)$ is bounded for $t \in \mathcal{T}_p$ as in (20). This completes the proof. ∎

Remark 6: Most results in the existing literature introduce specific time-varying gains for the prescribed-time control, such as $\mu(t) = (T/(T + t_0 - t))^m$ with $m \in \mathbb{N}_+$ in [21], $\mu(t) = \tan\left(\frac{\pi(t+t_0)}{2(T+t_0)}\right) + 1$ in [40], and $\mu(t) = \frac{\exp(c(T+t_0))-1}{\exp(c(T+t_0))-\exp(c(t+t_0))}$ with $c \in \mathbb{R}_+$ in [41]. In this paper, we introduce a general class of $\mathcal{K}_T$ functions in Definition 1 and incorporate them into $\mathcal{K}_\infty$ functions for the prescribed-time control in (12) and (13). Then, we investigate the conditions for $\mathcal{K}_T$ functions to achieve the prescribed-time convergence in Lemma 3 and, later on, establish a more general approach. ∎

B. Controller Design and Prescribed-Time Stability Analysis

For i-th agent, we denote

$$e_i = \begin{bmatrix} e_{i1}\\ e_{i2}\end{bmatrix} = \begin{bmatrix} q_i - \upsilon_0\\ r_i - S_0\upsilon_0\end{bmatrix},\ \ i \in \bar{\mathcal{V}} \tag{22}$$

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as the real tracking error of the i-th follower towards the leader. Note that e_i may not be available for the i-th agent since υ_0 and S_0 are only known to the neighbors of the leader system. Therefore, we denote

$$\hat{e}_i = \begin{bmatrix} \hat{e}_{i1} \\ \hat{e}_{i2} \end{bmatrix} = \begin{bmatrix} q_i - \upsilon_i \\ r_i - \dot{\upsilon}_i \end{bmatrix}, \ i \in \bar{\mathcal{V}} \tag{23}$$

as the local tracking error. Then, the relationship between e_i and $\hat{e}_i$ is

$$e_i = \hat{e}_i + \tilde{e}_i \tag{24}$$

where

$$\tilde{e}_i = \begin{bmatrix} \tilde{\upsilon}_i \\ z_i \end{bmatrix} = \begin{bmatrix} \upsilon_i - \upsilon_0 \\ \dot{\upsilon}_i - S_0\upsilon_0 \end{bmatrix}, \ i \in \bar{\mathcal{V}}. \tag{25}$$

Therefore, PTS is achieved if the prescribed-time convergence is achieved for $\hat{e}_i$ and $\tilde{e}_i$. Lemma 3 shows that the prescribed-time convergence is achieved for $\tilde{\upsilon}$. The next lemma shows that the prescribed-time convergence is achieved for z, where $z = \left[z_1^\top, \cdots, z_N^\top\right]^\top = \left[(\dot{\upsilon}_1 - S_0\upsilon_0)^\top, \cdots, (\dot{\upsilon}_N - S_0\upsilon_0)^\top\right]^\top$.

Lemma 4: Suppose conditions of Lemma 3 hold, and the functions γ_s, γ_υ satisfy (7) in Proposition 2 with $c_2/c_1 \leq \zeta/2$ and $\mu \in \mathcal{K}_T$. Then, the prescribed-time convergence is achieved for z, i.e., there exists a $\mathcal{KL}_T$ function $\tilde{\beta}_z$ such that

$$\|z(t)\| \leq \tilde{\beta}_z(\|\vartheta_z(t_0)\|, t)) = \alpha_z(\|\vartheta_z\|)\kappa^{-\frac{(1-\varepsilon)\zeta}{2}}(\gamma_\upsilon(\mu)) \tag{26}$$

where $\vartheta_z = \left[(\mathrm{vec}(\tilde{S}))^\top, \tilde{\upsilon}^\top\right]^\top$ and $\alpha_z \in \mathcal{K}_\infty$. Moreover, there exists a $\mathcal{KL}_T$ function $\tilde{\beta}_s$ such that

$$\|\dot{S}(t)\| \leq \tilde{\beta}_s(\|\tilde{S}(t_0)\|, t) = \tilde{\alpha}_s(\|\tilde{S}(t_0)\|)\kappa^{-\frac{(1-\varepsilon)\zeta}{2}}(\gamma_s(\mu)) \tag{27}$$

where $\tilde{\alpha}_s \in \mathcal{K}_\infty$. ∎

Proof: Due to (12) and (13), $\dot{S}$ and $\dot{\upsilon}$ can be described as follows:

$$\dot{S} = -\gamma_s(\mu)(H \otimes I_n)\tilde{S}, \tag{28}$$

$$\dot{\upsilon} = -\gamma_\upsilon(\mu)(H \otimes I_n)\tilde{\upsilon} + \Lambda_s\upsilon \tag{29}$$

where Λ_s is given in (16). Due to Lemma 3, (17) holds. Note that (17) and (28) imply that $\|\dot{S}\| \leq \|H \otimes I_n\|\tilde{\alpha}_s(\|\tilde{S}(t_0)\|)\gamma_s(\mu)\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu))$, then applying Proposition 2 leads to (27) with $\tilde{\alpha}_s(\|\tilde{S}(t_0)\|) = \gamma_s(\mu(t_0))\|H \otimes I_n\|\tilde{\alpha}_s(\|\tilde{S}(t_0)\|)$. Note that

$$\begin{aligned} z &= \dot{\upsilon} - \Lambda_s\upsilon + \Lambda_s\upsilon - 1_N \otimes S_0\upsilon_0 \\ &= \dot{\upsilon} - \Lambda_s\upsilon + \Lambda_{\tilde{s}}(1_N \otimes \upsilon_0) + \Lambda_s\tilde{\upsilon} \end{aligned} \tag{30}$$

where Λ_s and $\Lambda_{\tilde{s}}$ are given in (16). Note that (18) and (29) imply

$$\|\dot{\upsilon} - \Lambda_s\upsilon\| \leq \alpha_\upsilon(\|\tilde{\upsilon}(t_0)\|)\|(H \otimes I_n)\|\gamma_\upsilon(\mu)\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu)). \tag{31}$$

Since function γ_υ satisfies (7) with $c_1 = 1$, $c_2 = \zeta/2$, by Proposition 2, we obtain that $\gamma_\upsilon(\mu(t))\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu)) \leq \gamma_\upsilon(\mu(t_0))\kappa^{-\frac{(1-\varepsilon)\zeta}{2}}(\gamma_\upsilon(\mu))$. Due to (17), (18), (19), (30), (31), and the fact $\gamma_\upsilon(s) - \gamma_s(s) < 0$, one obtains (26) with some $\alpha_z \in \mathcal{K}_\infty$. ∎

Remark 7: There are many (in fact, an infinite number) of γ_s and γ_υ functions satisfying conditions in both Lemma 3 and 4. For example, given $\mu(t) = 1/(T + t_0 - t)$, one can find $\iota_1 = 2$, $\iota_2 = 3/2$, $w_1 = 1$, $w_2 = 2$ in Definition 1. γ_s and γ_υ can be chosen as $\gamma_s(\mu) = m_1\mu$, $\gamma_\upsilon(\mu) = m_2\mu$ with m_1, m_2 being finite constants satisfying $2/\zeta \leq c_1/c_2 < m_2 < m_1$. Since $c_2/c_1 \leq \zeta/2$, we can check $\frac{\mathrm{d}\gamma_s(s)}{\mathrm{d}s} = m_1 = \omega_1^{-1}\frac{c_2}{c_1}s^{-\iota_1}(\gamma_s(s))^2\frac{c_1}{m_1c_2} \leq \varepsilon\omega_1^{-1}\frac{c_2}{c_1}s^{-\iota_1}(\gamma_s(s))^2$ with $\varepsilon = c_1/(c_2m_1)$. Similarly, we can choose $\gamma_\upsilon(s)$ to satisfy Proposition 2, and Lemmas 3, 4. Note that when a different $\mu \in \mathcal{K}_T$ is used, it yields different ι_k, ω_k with $k = 1, 2$ and results in a different group of γ_s and γ_υ, as shown in Proposition 2 and 3. ∎

Combining Lemma 3 and 4 shows that the prescribed-time convergence is achieved for $\tilde{e}_i$ in (25), and it now suffices to prove that the prescribed-time convergence is achieved for $\hat{e}_i$ defined in (23). To this end, we introduce time-varying state transformation

$$\varpi_{i1} = \Gamma_\varpi(\mu)\hat{e}_{i1}, \tag{32}$$

$$\varpi_{i2} = \dot{\varpi}_{1i} = \dot{\Gamma}_\varpi(\mu)\hat{e}_{i1} + \Gamma_\varpi(\mu)\hat{e}_{i2} \tag{33}$$

where $\Gamma_\varpi(\mu(t)) = \gamma_\varpi(\mu(t))\tilde{K}$ with γ_ϖ being a twice-differentiable $\mathcal{K}_\infty$ function to be designed, and $\tilde{K} = \mathrm{diag}\{\tilde{k}_1, \cdots, \tilde{k}_n\} \in \mathbb{R}^{n\times n}$ with $\tilde{k}_i > 0$ for $i = 1, \cdots, n$. $\dot{\Gamma}_\varpi(\mu(t)) \in \mathbb{R}^{n\times n}$ is the time derivative of $\Gamma_\varpi(\mu(t))$.

For $i \in \bar{\mathcal{V}}$, the local prescribed-time tracking controllers are designed as

$$\xi_i = \varpi_{i2} + K_i\varpi_{i1}, \tag{34}$$

$$\begin{aligned} \tau_i &= C_i(q_i, r_i)r_i + G_i(q_i) \\ &\quad + M_i(q_i)\big(-(\rho_i + 2I_n)\xi_i - \eta_i + S_i^2\upsilon_i\big) \end{aligned} \tag{35}$$

where $K_i \in \mathbb{R}^{n\times n}$, $\rho_i = \mathrm{diag}\{\rho_{i1}, \cdots, \rho_{in}\} \in \mathbb{R}^{n\times n}$ with $\rho_{ij} > 0$ for $j = 1, \cdots, n$, and $\eta_i = \frac{\ddot{\Gamma}_\varpi(\mu)}{\Gamma_\varpi^2(\mu)}\varpi_{i1} - 2\frac{\dot{\Gamma}_\varpi^2(\mu)}{\Gamma_\varpi^3(\mu)}\varpi_{i1} + 2\frac{\dot{\Gamma}_\varpi(\mu)}{\Gamma_\varpi^2(\mu)}\varpi_{i2} + \frac{K_i}{\Gamma_\varpi(\mu)}\varpi_{i2}$, where $\ddot{\Gamma}_\varpi(\mu) \in \mathbb{R}^{n\times n}$ is the second-order derivative of $\Gamma_\varpi(\mu)$ with respect to t. We note that $(\Gamma_\varpi(\mu(t)))^l$ is positive definite for $l \in \mathbb{N}_+$ and $\forall t \in \mathcal{T}_p$, thus η_i can be computed for $t \in \mathcal{T}_p$. Then, we have the following theorem.

Theorem 1: Consider the closed-loop networked Euler-Lagrange systems composed of (1)-(2), distributed prescribed-time observers (12)-(13), and local prescribed-time tracking controllers (35) under Property 1 and Assumption 1, 2. Suppose the distributed prescribed-time observers (12)-(13) are designed according to Lemma 3 and 4. Let K_i in (34) be arbitrarily given. Suppose the $\mathcal{K}_\infty$ function γ_υ satisfies the conditions of Proposition 2 and 3 with $c_2/c_1 \leq c \leq \zeta/2$, and the $\mathcal{K}_\infty$ function γ_ϖ satisfies $\frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2} \geq 0$ and

$$\frac{\mathrm{d}\gamma_\varpi(s)}{\mathrm{d}s}/(s^{-\iota_1}\gamma_\varpi^{\iota_\varpi}(s)) < \infty, \tag{36}$$

$$\left(\frac{\mathrm{d}^2\gamma_\varpi(s)}{\mathrm{d}s^2}s^{\iota_1} + \frac{\mathrm{d}\gamma_\varpi(s)}{\mathrm{d}s}s^{\iota_1\iota_2}\right)/(\gamma_\varpi^2(s)) < \infty \tag{37}$$

for $\forall s \in [a_0, \infty)$ and $1 < \iota_\varpi < 3/2$. Then, the prescribed-time synchronization problem is solved and all internal signals in the closed-loop systems are bounded. ∎

Proof: The proof will be divided into three parts. We first show that ξ_i in (34) is bounded for $t \in \mathcal{T}_p$ which will be

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used to prove that ϖ_{i1} and ϖ_{i2} are bounded. Then, we prove that prescribed-time convergence is achieved for e_i. Last, the boundedness of the control signal is shown.

1) Boundedness of ξ_i: Taking the time derivative of ξ_i and upon using τ_i in (35) yields

$$\begin{aligned}\dot{\xi}_i &= \ddot{\Gamma}_\varpi(\mu)\hat{e}_{i1} + 2\dot{\Gamma}_\varpi(\mu)\hat{e}_{i2} + \Gamma_\varpi(\mu)\dot{\hat{e}}_{i2} + K_i\varpi_{i2}\\ &= \Gamma_\varpi(\mu)\Big(M_i^{-1}(q_i)\big(\tau_i - d_i - C_i(q_i, r_i)r_i\\ &\quad - G_i(q_i)\big) - \ddot{\upsilon}_i + \eta_i\Big)\\ &= \Gamma_\varpi(\mu)\big(-(\rho_i + 2I_n)\xi_i - \varsigma_i - M_i^{-1}(q_i)d_i\big) \qquad (38)\end{aligned}$$

where

$$\varsigma_i = \ddot{\upsilon}_i - S_i^2\upsilon_i. \qquad (39)$$

Define the Lyapunov function candidate for ξ_i-dynamics in (38) as $W_i(\xi_i) = \frac{1}{2}\xi_i^\top\xi_i$. Using (38) and (39), the time derivative of $W_i(\xi_i)$ is

$$\dot{W}_i(\xi_i) = \xi_i^\top\Gamma_\varpi(\mu)\big(-(\rho_i + 2I_n)\xi_i - \varsigma_i - M_i^{-1}(q_i)d_i\big). \qquad (40)$$

By Property 1, Assumption 1 and Young's inequality, one has $-\xi_i^\top\Gamma_\varpi(\mu)M_i^{-1}(q_i)d_i \le \gamma_\varpi(\mu)\|\xi_i\|\|\tilde{K}\|\|M_i^{-1}(q_i)\|d_i\| \le \gamma_\varpi(\mu)\underline{k}_n^{-1}\bar{d}_i\tilde{k}_{\max}\|\xi_i\| \le \xi_i^\top\Gamma_\varpi(\mu)\xi_i + \gamma_\varpi(\mu)\bar{d}_i^2\tilde{k}_{\max}^2/(4\underline{k}_n^2\tilde{k}_{\min})$, where $\tilde{k}_{\min} = \min\{\tilde{k}_1, \cdots, \tilde{k}_n\}$, $\tilde{k}_{\max} = \max\{\tilde{k}_1, \cdots, \tilde{k}_n\}$ and we used $\|\tilde{K}\| = \tilde{k}_{\max}$.

Since $\Lambda_s\upsilon - 1_N \otimes S_0\upsilon_0 = \Lambda_s\upsilon - \Lambda_s(1_N \otimes \upsilon_0) + \Lambda_s(1_N \otimes \upsilon_0) - 1_N \otimes S_0\upsilon_0 = \Lambda_s\tilde{\upsilon} + \Lambda_{\tilde{s}}(1_N \otimes \upsilon_0)$. Then by (29), $\ddot{\upsilon}$ can be expressed as

$$\begin{aligned}\ddot{\upsilon} &= -\dot{\gamma}_\upsilon(\mu)\bar{H}\tilde{\upsilon} - \gamma_\upsilon(\mu)\bar{H}\dot{\tilde{\upsilon}} + \Lambda_{\dot{s}}\upsilon + \Lambda_s\dot{\upsilon}\\ &= \Phi(\mu)\tilde{\upsilon} - \gamma_\upsilon(\mu(t))\bar{H}\Lambda_{\tilde{s}}\bar{\upsilon}_0 + \Lambda_{\dot{s}}\upsilon + \Lambda_s^2\upsilon \qquad (41)\end{aligned}$$

where $\Phi(\mu) = -\dot{\gamma}_\upsilon(\mu)\bar{H} + \gamma_\upsilon^2(\mu)\bar{H}^2 - \gamma_\upsilon(\mu)\bar{H}\Lambda_s - \gamma_\upsilon(\mu)\Lambda_s\bar{H}$, $\Lambda_{\dot{s}} = \mathrm{diag}\{\dot{S}_1, \cdots, \dot{S}_n\}$, $\bar{H} = H \otimes I_n$ and $\bar{\upsilon}_0 = 1_N \otimes \upsilon_0$. Upon using (17), (18), (19) and (20), the terms in (41) satisfy $\|\Phi(\mu)\tilde{\upsilon}\| \le \epsilon_\upsilon\bar{\gamma}_\upsilon(\mu)\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu))$, $\gamma_\upsilon(\mu(t))\bar{H}\Lambda_{\tilde{s}}\bar{\upsilon}_0 \le \epsilon_s\gamma_s(\mu)\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu))$ and $\Lambda_{\dot{s}}\upsilon \le \epsilon_s'\gamma_s(\mu)\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu))$, where $\bar{\gamma}_\upsilon(t)$ is defined in (8), $\epsilon_\upsilon = \max\{\|\bar{H}\|, \|\bar{H}\|^2, 2\bar{s}\|\bar{H}\|\}\alpha_\upsilon(\|\tilde{\upsilon}(t_0)\|)$, $\epsilon_s = \|\bar{H}\|\sup_{t\in\mathcal{T}_p}\|\bar{\upsilon}_0(t)\|\alpha_s(\|\tilde{S}(t_0)\|)$ and $\epsilon_s' = \bar{\upsilon}\|\bar{H}\|\tilde{\alpha}_s(\|\tilde{S}(t_0)\|)$.

Let $\varsigma = [\varsigma_1^\top, \cdots, \varsigma_N^\top]^\top$ with ς_i denoted in (39). Using (41) and the above inequalities leads to $\|\varsigma\| \le \epsilon_\upsilon\bar{\gamma}_\upsilon(t)\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu)) + (\epsilon_s + \epsilon_s')\gamma_s(\mu(t))\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu))$. By Proposition 3 with $c_2/c_1 \le c \le \zeta/2$, $\bar{\gamma}_\upsilon(t)\kappa^{-\frac{\zeta}{2}}(\gamma_\upsilon(\mu))$ and $\gamma_s(\mu)\kappa^{-\frac{\zeta}{2}}(\gamma_s(\mu))$ are $\mathcal{D}_T$ functions. Therefore, ς satisfies

$$\|\varsigma\| \le \beta_\varsigma(\|\vartheta_z(t_0)\|, t) \qquad (42)$$

for some $\mathcal{KL}_T$ functions β_ς, where ϑ_z is defined in 4. Then by (42), $\|\varsigma\| \le \phi = \beta_\varsigma(\|\vartheta_z(t_0)\|$, where ϕ is some positive constant. As a result, the bound of $-\xi_i^\top\gamma_\varpi(\mu(t))\varsigma_i$ is calculated as

$$-\xi_i^\top\Gamma_\varpi(\mu)\varsigma_i \le \xi_i^\top\Gamma_\varpi(\mu)\xi_i + \gamma_\varpi(\mu)\|\phi\|^2\tilde{k}_{\max}^2/(4\tilde{k}_{\min}) \qquad (43)$$

where we used the fact $\|\varsigma_i\| \le \|\varsigma\|$. By (40) and (43), $\dot{W}_i(\xi_i)$ satisfies $\dot{W}_i(\xi_i) \le -\xi_i^\top\Gamma_\varpi(\mu)\rho_i\xi_i + \gamma_\varpi(\mu)\Delta_i \le -\varrho_i\gamma_\varpi(\mu)W_i(\xi_i) + \gamma_\varpi(\mu)\Delta_i$, where $\varrho_i = 2\min\{\rho_{i1}\tilde{k}_1, \cdots, \rho_{in}\tilde{k}_n\}$ and $\Delta_i = \bar{d}_i^2\tilde{k}_{\max}^2/(4\underline{k}_n^2\tilde{k}_{\min}) + \|\phi\|^2\tilde{k}_{\max}^2/(4\tilde{k}_{\min})$. Applying Lemma 2 gives

$$\|\xi_i(t)\| \le \bar{\xi}_i = \alpha_{\xi_i}(\|\xi_i(t_0)\|) \qquad (44)$$

for some $\mathcal{K}_\infty^e$ function α_{ξ_i}.

2) Prescribed-Time Convergence of e_i: We first prove the boundedness of ϖ_{i1} and ϖ_{i2}. Define the Lyapunov candidate function $J_i(\varpi_{i1}) = \varpi_{i1}^\top\varpi_{i1}$. Taking the time derivative of $J_i(\varpi_{i1})$ along (33) yields

$$\dot{J}_i(\varpi_{i1}) = 2\varpi_{i1}^\top(-K_i\varpi_{i1} + \xi_i) \le c_iJ_i(\varpi_{i1}) + \bar{\xi}_i^2 \qquad (45)$$

where $c_i = 2\|K_i\| + 1 > 0$, $\bar{\xi}_i$ is defined in (44) and we used $\dot{\varpi}_{i1} = \varpi_{i2}$. For $t \in \mathcal{T}_p$, applying the comparison lemma to (45) yields

$$\begin{aligned}&J_i(\varpi_{i1}(t))\\ &\le \exp\big(c_i(t - t_0)\big)J_i(\varpi_{i1}(t_0)) + \int_{t_0}^t \exp\big(c_i(t - s)\big)\bar{\xi}_i^2\mathrm{d}s\\ &\le \exp\big(c_i(t - t_0)\big)(J_i(\varpi_{i1}(t_0)) + \bar{\xi}_i^2c_i^{-1}) - \bar{\xi}_i^2c_i^{-1}\\ &\le \exp\big(c_iT\big)(J_i(\varpi_{i1}(t_0)) + \bar{\xi}_i^2c_i^{-1}). \qquad (46)\end{aligned}$$

Therefore, $J_i(\varpi_{i1}(t))$, and thus $\varpi_{i1}(t)$, are bounded for $t \in \mathcal{T}_p$. By (46), the upper bound $\bar{\varpi}_{i1}$ of $\varpi_{i1}(t)$ can be expressed as $\|\varpi_{i1}(t)\| \le \bar{\varpi}_{i1} = \exp\big(\omega_iT/2\big)(\|\varpi_{i1}(t_0)\| + \bar{\xi}_ic_i^{-\frac{1}{2}})$.

Define $\varpi_i = [\varpi_{i1}^\top, \varpi_{i2}^\top]^\top$. By (32) and (33), we have $\hat{e}_i = \psi(\mu)\varpi_i$, where

$$\psi(\mu) = \begin{bmatrix}(1/\gamma_\varpi(\mu)) & 0\\ -(\dot{\gamma}_\varpi(\mu)/\gamma_\varpi^2(\mu)) & (1/\gamma_\varpi(\mu))\end{bmatrix} \otimes \tilde{K}^{-1}.$$

From (36), one has $\sup_{s\in[a_0,\infty)}\Big[\frac{\mathrm{d}\gamma_\varpi(s)}{\mathrm{d}s}/(s^{-\iota_1}\gamma_\varpi^{\iota_\varpi}(s))\Big] = g_1 < \infty$. Since $1 < \iota_\varpi < 3/2$, there exists a $0 < p_1 < 1/2$ such that $\iota_\varpi + p_1 = 3/2$. By (3) and (36), $\dot{\gamma}_\varpi(\mu)$ satisfies

$$\begin{aligned}\dot{\gamma}_\varpi(\mu) &= \frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu}\dot{\mu}\\ &\le g_1\mu^{-\iota_1}\gamma_\varpi^{\frac{3}{2}-p_1}(\mu)\dot{\mu} \le g_1w_1\gamma_\varpi^{\frac{3}{2}-p_1}(\mu) \qquad (47)\end{aligned}$$

where we used the facts that $\frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu} \ge 0$ for $\mu \in [a_0, \infty)$ due to $\gamma_\varpi \in \mathcal{K}_\infty$, and $\dot{\mu}(t) > 0$ for $t \in \mathcal{T}_p$ since $\mu(t)$ is strictly increasing. Define $\Psi(\mu) = \|\psi(\mu)\|$. Then by (47) and the fact $\|A \otimes B\| = \|A\|\|B\|$, one has $\Psi(\mu) \le \tilde{k}_{\min}^{-1}\big(1/\gamma_\varpi(\mu) + \dot{\gamma}_\varpi(\mu)/\gamma_\varpi^2(\mu)\big) \le \tilde{k}_{\min}^{-1}\big((\gamma_\varpi(\mu))^{-1} + g_1\omega_1(\gamma_\varpi(\mu))^{-\frac{1}{2}-p_1}\big) = g_1'(\gamma_\varpi(\mu))^{-p_1-\frac{1}{2}}$, where $\tilde{k}_{\min} = \min\{\tilde{k}_1, \cdots, \tilde{k}_n\}$ and $g' = (g_1\omega_1 + (\gamma_\varpi(\mu(t_0))^{\frac{1}{2}-p_1})\tilde{k}_{\min}^{-1}$. Let $\hat{e} = [\hat{e}_1^\top, \cdots, \hat{e}_N^\top]^\top$. Therefore, $\hat{e}$ satisfies

$$\begin{aligned}\|\hat{e}(t)\| &\le \|\psi(\mu)\|\|\varpi\|\\ &\le \Psi(\mu(t))\bar{\varpi} \le \hat{\beta}_e(\bar{\varpi}, t) = \alpha_\varpi(\bar{\varpi})\gamma_\varpi^{-p_1-\frac{1}{2}}(\mu) \qquad (48)\end{aligned}$$

with some $\mathcal{KL}_T$ function $\hat{\beta}_e$, where $\bar{\varpi} = \bar{\varpi}_1 + \bar{\varpi}_2$ and $\alpha_\varpi(\bar{\varpi}) = g_1'\bar{\varpi}$. Inequality (48) means that $\hat{e}_i(t)$ is bounded for $t \in \mathcal{T}_p$ and converges to zero as $t \to t_0 + T$. Due to (24) and Lemmas 3 and 4, using (48) and the fact $\gamma_\upsilon(s) - \gamma_s(s) < 0$, the bound of e is obtained as

$$\|e(t)\| \le \beta_\upsilon(\|\tilde{\upsilon}(t_0)\|, t) + \tilde{\beta}_z(\|\vartheta_z(t_0)\|, t)) + \beta_e(\bar{\varpi}, t)$$

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$$\leq \alpha_e(\vartheta_e(t_0)) \max\left\{\gamma_\varpi^{-p_1-\frac{1}{2}}(\mu), \kappa^{-(1-\varepsilon)\zeta/2}(\gamma_\upsilon(\mu))\right\}$$
$$= \beta_e(\|\vartheta_e(t_0)\|, t) \tag{49}$$

with some $\mathcal{KL}_T$ function β_e where $\vartheta_e = [\mathrm{vec}(\tilde{S})^\top, \tilde{\upsilon}^\top, \bar{\varpi}^\top]^\top$. The prescribed-time convergence is achieved for e_i. In other words, $e_i(t)$ is bounded for $t \in \mathcal{T}_p$ and converges to zero as $t \to t_0 + T$. As a result, the prescribed-time synchronization between leader and followers is achieved.

3) Boundedness of the Control Signal τ_i: By (37), we have $\ddot{\gamma}_\varpi(\mu) = \frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2}\dot{\mu}(t) + \frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu}\ddot{\mu}(t) \leq \frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2}w_1\mu^{\iota_1} + \frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu}w_2 w_1^{\iota_1}\mu^{\iota_1\iota_2} \leq \bar{w}\left(\frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2}\mu^{\iota_1} + \frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu}\mu^{\iota_1\iota_2}\right) \leq \bar{w}g_2\gamma_\varpi^2(\mu)$ for some finite number $g_2 > 0$, where $\bar{w} = \max\{w_1, w_2 w_1^{\iota_1}\}$. By (47), η_i satisfies $\|\eta_i\| \leq \bar{w}g_2\bar{\varpi}_{i1} + 2(g_1w_1)^2(\gamma_\varpi(\mu))^{-1}\bar{\varpi}_{i1} + 2g_1w_1(\gamma_\varpi(\mu))^{-1}\bar{\varpi}_{i2} + \|K_i\|(\gamma_\varpi(\mu))^{-1}\bar{\varpi}_{i2} \leq (\bar{w}g_2\gamma_\varpi(a) + 2(g_1w_1)^2 + 2g_1w_1 + \|K_i\|)\bar{\varpi}_i(\gamma_\varpi(a_0))^{-1}$, where we used the fact $\gamma_\varpi(\mu) \geq \gamma_\varpi(a_0)$.

Note that $\dot{q}_i$ in (1) satisfies $\|\dot{q}_i\| \leq \|e_i\| + \|S_0\upsilon_0\| \leq \beta_e(\|r_e(t_0)\|, t_0) + \|S_0\|\bar{\upsilon}_0$. By (19) and (18), $S_i^2\upsilon_i$ is also bounded. Thus, we can conclude that τ_i, $i = 1, \cdots, N$ is bounded. ■

Remark 8: Let $\mu(t) = (\frac{1}{T+t_0-t})^{m_1}$ and $\gamma_\varpi(\mu) = \mu^{m_2}$. As a result, $w_1 = m_1$, $\iota_1 = \frac{m_1+1}{m_1}$, $w_2 = m_1+1$, $\iota_2 = \frac{m_1+2}{m_1+1}$, $\frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu} = m_2\mu^{m_2-1}$ and $\frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2} = m_2(m_2-1)\mu^{m_2-2}$. Let us choose $m_1 > 0$, $m_2 > 2(m_1+1)/m_1+2$ and $m_1m_2 > 2$. Then, (36) and (37) are satisfied. ▮

Remark 9: For the closed-loop networked Euler-Lagrange systems, there are three incorporations of class $\mathcal{K}_T$ and comparison functions: $\gamma_s(\mu)$, $\gamma_\upsilon(\mu)$, and $\gamma_\varpi(\mu)$. The function $\gamma_s(\mu)$ primarily provides a prescribed-time convergence gain for the S_i-dynamics. Meanwhile, $\gamma_\upsilon(\mu)$ offers a prescribed-time convergence gain for the υ_i-dynamics. The function $\gamma_\varpi(\mu)$ is directly fed back into the local controller, furnishing a prescribed-time convergence gain for the local tracking error $\hat{e}_i$. Due to the decomposition of the synchronization error as described in (23), (24), and (25), as shown in (49), the prescribed-time convergence of the synchronization error e_i is influenced by the three gains: $\gamma_s(\mu)$, $\gamma_\upsilon(\mu)$, and $\gamma_\varpi(\mu)$. Specifically, the faster μ increases with respect to time within $\mathcal{T}_p$, or the faster $\gamma_s(\mu)$, $\gamma_\upsilon(\mu)$, and $\gamma_\varpi(\mu)$ increase with respect to μ, the quicker the synchronization error converges in prescribed time. ▮

Remark 10: Adaptive fuzzy prescribed-time control [29] and adaptive neural prescribed-time control [42] employ approximations with fuzzy logic or ANNs within compact sets, resulting in time-varying gains capped by upper bounds, thus preventing unbounded growth. Consequently, these controls achieve only practical prescribed-time convergence. Furthermore, in contrast to robust finite-time control [43] and robust fixed-time control [14], [15], our robust prescribed-time synchronization purely utilizes time-varying gains to counteract the impact of external disturbances on convergence. It does not involve any compensation measures within the controller, nor does it employ any disturbance information. ▮

V. Simulation and Experimental Results

A. Numerical Simulation

Consider a group of six robotic manipulators given by (1) where $q_i = [q_{i1}, q_{i2}]^\top \in \mathbb{R}^2$ and

$$M_i(q_i) = \begin{bmatrix} \theta_{i1} + \theta_{i2} + 2\theta_{i3}\cos(q_{i2}) & \theta_{i2} + \theta_{i3}\cos(q_{i2}) \\ \theta_{i2} + \theta_{i3}\cos(q_{i2}) & \theta_{i4} \end{bmatrix},$$
$$C_i(q_i, r_i) = \begin{bmatrix} -\theta_{i3}\sin(q_{i2})r_{i1} & -2\theta_{i3}\sin(q_{i2})r_{i1} \\ 0 & \theta_{i3}\sin(q_{i2})r_{i2} \end{bmatrix},$$
$$G_i(q_i) = \begin{bmatrix} \theta_{i5}g\cos(q_{i1}) + \theta_{i6}g\cos(q_{i1}+q_{i2}) \\ \theta_{i6}g\cos(q_{i1}+q_{i2}) \end{bmatrix}$$

where $\theta_{i1} = 6\mathrm{kg}\cdot\mathrm{m}^2$, $\theta_{i2} = 0.8\mathrm{kg}\cdot\mathrm{m}^2$, $\theta_{i3} = 0.1\mathrm{kg}\cdot\mathrm{m}^2$, $\theta_{i4} = 5.5\mathrm{kg}\cdot\mathrm{m}^2$, $\theta_{i5} = 0.1\mathrm{kg}\cdot\mathrm{m}$, $\theta_{i6} = 0.1\mathrm{kg}\cdot\mathrm{m}$ for $i = 1, \cdots, 6$, and $g = 9.8\mathrm{m}\cdot\mathrm{s}^2$. The external disturbance is denoted as $d_i = 1_2 \otimes \delta(t), i = 1, \cdots, 6$, where $\delta(t)$ is impulse noise with sparsity 0.01 and pulse peak 5.

The leader system is given by (2) with $S_0 = [0, -1; -1, 0]$ and $\upsilon_0(t_0) = [0.5, 0.5]^\top$. The information flow among all the subsystems and the leader is described by the digraph as $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 6$, which contains a spanning tree with node 0 as the root, and thus Assumption 2 is satisfied. We select $Q = I_{12}$ and then $\zeta = \underline{\lambda}_Q\bar{\lambda}_P^{-1} = 0.3406$.

The prescribed-time function $\mu(t)$ is chosen as $\mu(t) = (1/(T+t_0-t))^2$ with $t_0 = 0$ and prescribed scalar $T = 1s$. Thus, $\mu(t)$ satisfies Definition 1 with $\iota_1 = \frac{3}{2}$, $\iota_2 = \frac{4}{3}$, $w_1 = 2$ and $w_2 = 3$. The observer gains in (12), (13) are $\gamma_s(\mu) = 16\mu$, $\gamma_\upsilon(\mu) = 14\mu$. We check $\gamma_s(\mu) > \gamma_\upsilon(\mu)$, $\omega_1\frac{\mathrm{d}\gamma_s(\mu)}{\mathrm{d}\mu}\mu^{\iota_1}(\gamma_s(\mu))^{-2} \leq (T+t_0-t)/8 \leq 1/8$, $\omega_1\frac{\mathrm{d}\gamma_\upsilon(\mu)}{\mathrm{d}\mu}\mu^{\iota_1}(\gamma_\upsilon(\mu))^{-2} \leq (T+t_0-t)/7 \leq 1/7$, $\frac{\mathrm{d}^2\gamma_\upsilon(\mu)}{\mathrm{d}\mu^2}\gamma_\upsilon(\mu) + \frac{\mathrm{d}\gamma_\upsilon(\mu)}{\mathrm{d}\mu}w_2w_1^{\iota_2-1}\mu^{\iota_1(\iota_2-1)} \leq 0.27\frac{\mathrm{d}\gamma_\upsilon(\mu)}{\mathrm{d}\mu}\gamma_\upsilon(\mu)$, thus conditions of Lemma 3 and 4 are satisfied. The time-varying state transformation function $\gamma_\varpi(\mu)$ in (32) is $\tilde{K} = I_2$ and $\gamma_\varpi(\mu) = \mu^2$, and then $\frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu} = 2\mu$, $\frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2} = 2$. We check $\frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2}\mu^{\iota_1}(\gamma_\varpi(\mu))^{-\frac{3}{2}} = 2(\gamma_\varpi(\mu))^{-\frac{1}{4}}$, $\frac{\mathrm{d}^2\gamma_\varpi(\mu)}{\mathrm{d}\mu^2}\mu^{\iota_1} + \frac{\mathrm{d}\gamma_\varpi(\mu)}{\mathrm{d}\mu}\mu^{\iota_1\iota_2} = 2\mu^{\frac{3}{2}} + 2\mu^3 \leq 2\gamma_\varpi^2(\mu)$, thus (36) and (37) are satisfied.

The initial values are $S_i = [1, 1; 1, 1]$, $\upsilon_i = [0, 0]^\top$, $q_i = [0.5, 0.5]^\top$, $r_i = [2, 2]^\top$, $i = 1, \cdots, 6$, and control parameters are selected as $\rho_i = 3I_2$, $K_i = I_2$, $i = 1, \cdots, 6$. The sampling time for ordinary differential equation solvers tool is $5\times10^{-5}s$. The simulation results are presented in Fig. 1-4, demonstrating that the distributed estimate errors, synchronization position errors, and synchronization velocity errors remain bounded for $t \in \mathcal{T}_p$ and converge to zero as $t \to T + t_0$. Additionally, the uniform boundedness of the control input τ_i is illustrated in Fig. 4. Furthermore, we verify that the finite settling time is independent of initial values and control parameters, as shown in Fig. 5. Let $e_p = [e_{11}^\top, \cdots, e_{N1}^\top]^\top$ and $e_v = [e_{12}^\top, \cdots, e_{N2}^\top]^\top$ be the position error and velocity error, where e_{i1} and e_{i2} are denoted in (22). The trajectories of e_p and e_v subject to different prescribed times are shown in Fig. 6, demonstrating the PTS algorithm's robustness and its ability to be specified *a priori*.

To further demonstrate the advantages of our proposed PTS algorithm, a comparison with fixed-time

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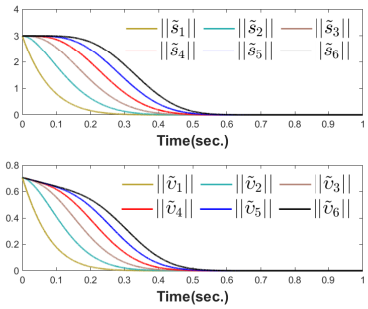

Fig. 1. The norm of distributed estimate errors $\tilde{s}_i$ and $\tilde{\upsilon}_i$ for $i = 1, \cdots, 6$.

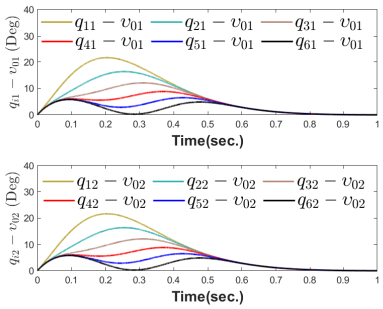

Fig. 2. Synchronization velocity errors $r_i - \dot{\upsilon}_0$, $i = 1, \cdots, 6$.

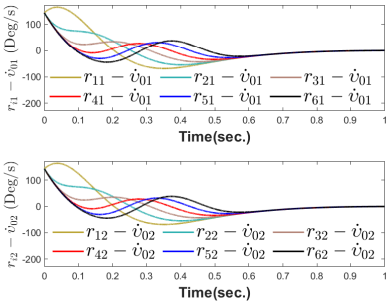

Fig. 3. Synchronization velocity errors $r_i - \dot{\upsilon}_0$, $i = 1, \cdots, 6$.

synchronization from [13] and [14] is carried out. The distributed fixed-time observers and local fixed-time tracking controllers are designed as follows, for $i = 1, \cdots, 6$, $\dot{S}_i = -c_1\chi_i - c_2\mathrm{sig}^{\beta_1}(\chi_i) - c_3\mathrm{sig}^{\alpha_1}(\chi_i)$, $\dot{\upsilon}_i = S_i\upsilon_i - c_1\tilde{\chi}_i - c_2\mathrm{sig}^{\beta_1}(\tilde{\chi}_i) - c_3\mathrm{sig}^{\alpha_1}(\tilde{\chi}_i)$, $\tilde{s}_i = \hat{e}_{i2} + \gamma_1\big(\mathrm{sig}^{\alpha_1}(\hat{e}_{i1}) + \mathrm{sig}^{\beta_1}(\hat{e}_{i1})\big)$ and $\tau_i = C_i(q_i, r_i)r_i + G_i(q_i) - \gamma_2 M_i(q_i)\big(\mathrm{sig}^{\alpha_2}(\tilde{s}_i) + \mathrm{sig}^{\beta_2}(\tilde{s}_i)\big) - \gamma_1 M_i(q_i)(\alpha_1\mathrm{diag}\{|\hat{e}_{i1}|^{\alpha_1-1}\}\hat{e}_{i2} + \beta_1\mathrm{diag}\{|\hat{e}_{i1}|^{\beta_1-1}\}\hat{e}_{i2})$, where $\chi_i = \sum_{j\in\mathcal{N}_i}(S_i - S_j)$, $\tilde{\chi}_i = \sum_{j\in\mathcal{N}_i}(\upsilon_i - \upsilon_j)$, $\hat{e}_{i1}$ and $\hat{e}_{i2}$ are denoted in (23), $\mathrm{sig}^{\beta_1}(\chi_i) = [\mathrm{sign}(\chi_{i1})|\chi_{i1}|^{\beta_1}, \mathrm{sign}(\chi_{i2})|\chi_{i2}|^{\beta_2}]^\top$ with similar variables represented in the same manner. The design parameters are $c_1 = 8.4$, $c_2 = 1$, $c_3 = 1$, $\alpha_1 = \alpha_2 = 3$, $\beta_1 = \beta_2 = 3/5$, $\gamma_1 = \gamma_2 = 2$. Based on [13], [14], the upper bound of the settling-time T is $T \leq \frac{2}{a_1(\alpha_2-1)} + \frac{2}{b_1(1-\beta_2)} = 2.94s$, where $a_1 = 2^{[(1+\alpha_2)/2]}(Nn)^{[(1-\alpha_2)/2]}\gamma_2$ and $b_1 = 2^{[(1+\beta_2)/2]}\gamma_2$ with $N = 6$ and $n = 2$ denoting the number of followers and the dimension of the leader system, respectively.

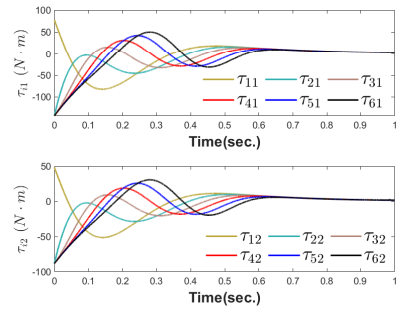

Fig. 4. The trajectories for the control input $\tau_i, i = 1, \cdots, 6$.

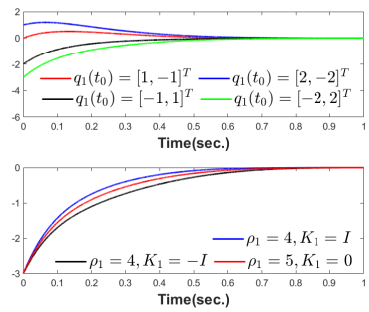

Fig. 5. Trajectories of $q_{11} - \upsilon_{01}$ subject to different initial values and different control parameters.

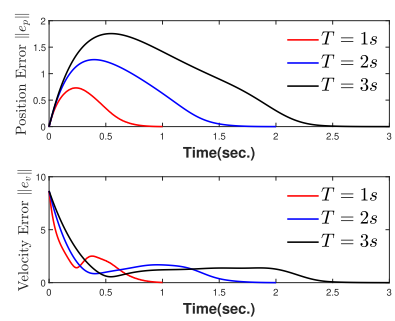

Fig. 6. Trajectories of position error e_p and velocity error e_v subject to different prescribed times T.

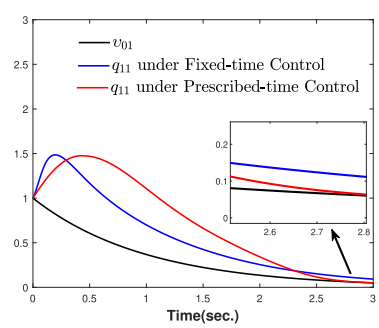

Fig. 7. The trajectories of q_{11} under different control algorithms.

The sampling time for the ordinary differential equation solver is $5 \times 10^{-5}s$, and the total simulation time for our PTS algorithm and the fixed-time synchronization algorithm is $3s$. The simulation results are given in Fig. 7-9. As shown

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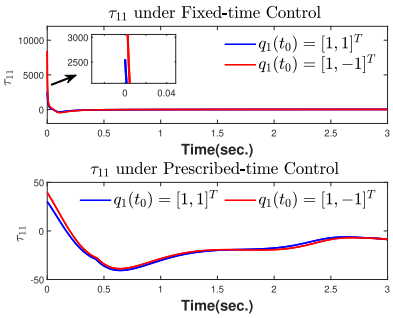

Fig. 8. Trajectories of torque input τ_{11} under different control algorithms and initial values.

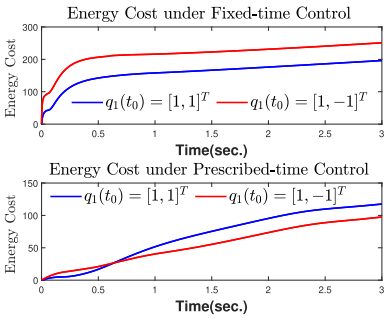

Fig. 9. Energy cost under different control algorithms and initial values.

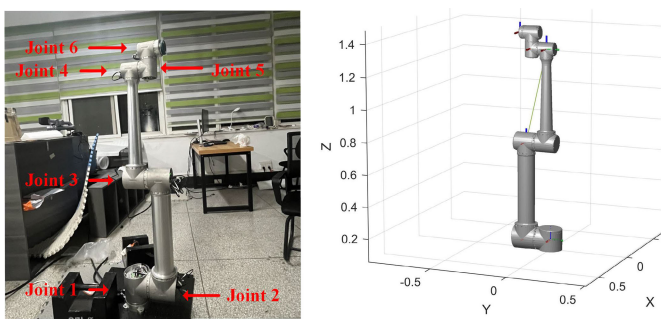

Fig. 10. The robot image and model diagram of the 6-DOF manipulator.

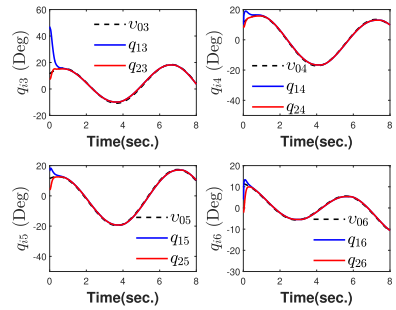

Fig. 11. The experimental and desired joint trajectories of the manipulators.

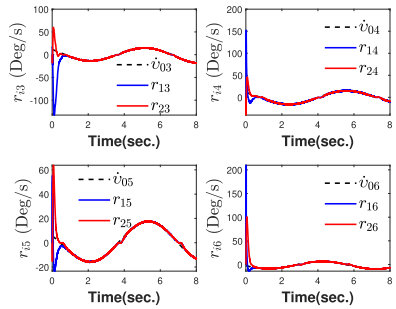

Fig. 12. The experimental and desired joint angular velocity of the manipulators.

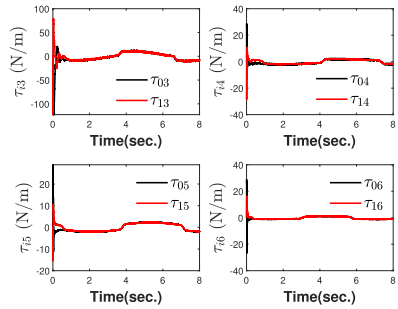

Fig. 13. The experimental torque outputs of the manipulators.

in Fig. 7, the convergence accuracy of our proposed PTS algorithm is better in the presence of external disturbances. The first control torque τ_{11} of follower 1 under different control schemes is displayed in Fig. 8, which demonstrates that the amplitude of the control torque under the PTS algorithm is lower. The energy cost $\mathrm{EC} = \int_{t_0}^{t} \|\tau_{11}(s)\|_1 \mathrm{d}s$ by control torque τ_{11} is shown in Fig. 9, indicating that under the same energy measurement standard, the PTS algorithm requires less energy. The trajectories of the other followers are similar and thus not repeated here.

B. Real-Robot Experiment

To verify the effectiveness of the proposed algorithm from the perspective of practical implementation, experimental studies are performed in this subsection using the 6-DOF industrial manipulator shown in Fig.10.

In this experiment, we set two manipulators as the followers and the communication graph is $0 \to 1 \to 2$. The leader system in (2) is designed as $S_0 = [0, -0.04, -0.33, -0.6, 0.45, 0.1; 0.04, 0, -0.4, -0.56, -0.6, -0.3; 0.33, 0.4, 0, -0.18, -0.06, 0.5; 0.6, 0.56, 0.18, 0, 0.1, -0.47; -0.45, 0.6, 0.06, -0.1, 0, 0.39; -0.1, 0.3, -0.5, 0.47, -0.39, 0]$ and $\upsilon_0(t_0) = [0.2, 0.2, 0.2, 0.2, 0.2, 0.2]^\top$. By adjusting the values of S_0 and $\upsilon_0(t_0)$, the leader system can generate desired trajectories for the manipulators to execute a place task as in [44]. The prescribed-time function $\mu(t)$ is chosen as $\mu(t) = (T + t_0)/(T + t_0 - t) + 5$ with $t_0 = 0$ and prescribed-time scalar $T = 8s$. Thus, μ is a $\mathcal{K}_T$ function in Definition 1 with $\iota_1 = 2$, $\iota_2 = 3/2$, $\omega_1 = 1/(T + t_0)$ and $\omega_2 = 2(T + t_0)^{-1/2}$. The observer gains in (12) and (13) are $\gamma_s(\mu) = 5\mu$ and $\gamma_\upsilon(\mu) = 4\mu$. We check $\gamma_s(\mu) > \gamma_\upsilon(\mu)$, $\omega_1 \frac{\mathrm{d}\gamma_s(\mu)}{\mathrm{d}\mu} \mu^{\iota_1} (\gamma_s(\mu))^{-2} = 1/(5(T + t_0))$, $\omega_1 \frac{\mathrm{d}\gamma_\upsilon(\mu)}{\mathrm{d}\mu} \mu^{\iota_1} (\gamma_\upsilon(\mu))^{-2} = 1/(4(T + t_0))$, $\frac{\mathrm{d}^2\gamma_\upsilon(\mu)}{\mathrm{d}\mu^2} \gamma_\upsilon(\mu) + \frac{\mathrm{d}\gamma_\upsilon(\mu)}{\mathrm{d}\mu} w_2 w_1^{\iota_2 - 1} \mu^{\iota_1(\iota_2 - 1)} \leq 2^{-1}(T + t_0)^{-3/2} \frac{\mathrm{d}\gamma_\upsilon(\mu)}{\mathrm{d}\mu} \gamma_\upsilon(\mu)$, and thus conditions of Lemma 3 and 4 are satisfied.

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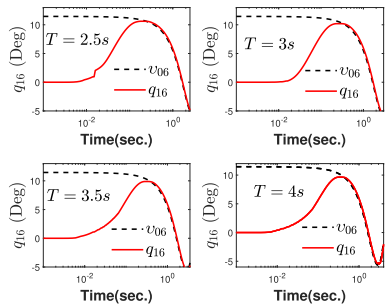

Fig. 14. The experimental and desired trajectories of the 6-th joint's position subject to different prescribed times T.

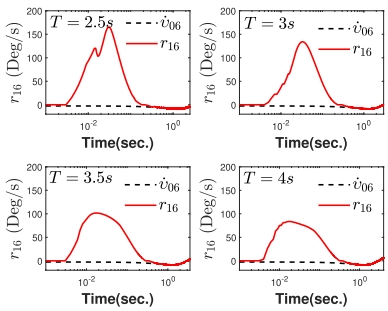

Fig. 15. The experimental and desired trajectories of the 6-th joint's velocity subject to different prescribed times T.

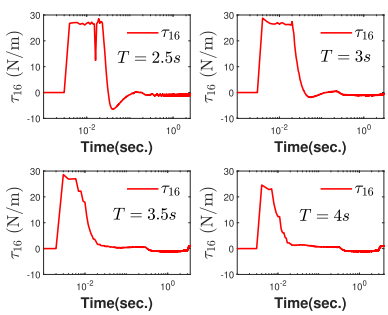

Fig. 16. The experimental and desired trajectories of the 6-th joint's torque subject to different prescribed times T.

The time-varying state transformation function $\gamma_{\varpi}(\mu)$ in (32) is given by $\tilde{K} = \text{diag}\{8, 8, 10, 10, 5, 5\}$ and $\gamma_{\varpi}(\mu) = \mu^2$. Then $\frac{\mathrm{d}\gamma_{\varpi}(\mu)}{\mathrm{d}\mu} = 2\mu$, $\frac{\mathrm{d}^2\gamma_{\varpi}(\mu)}{\mathrm{d}\mu^2} = 2$. We check $\frac{\mathrm{d}^2\gamma_{\varpi}(\mu)}{\mathrm{d}\mu^2}\mu^{\iota_1}(\gamma_{\varpi}(\mu))^{-\frac{3}{2}} = 2(\gamma_{\varpi}(\mu))^{-\frac{1}{2}}$, $\frac{\mathrm{d}^2\gamma_{\varpi}(\mu)}{\mathrm{d}\mu^2}\mu^{\iota_1} + \frac{\mathrm{d}\gamma_{\varpi}(\mu)}{\mathrm{d}\mu}\mu^{\iota_1\iota_2} = 2\mu^2 + 2\mu^4 \leq 4\gamma_{\varpi}^2(\mu)$, and thus (36) and (37) are satisfied.

The initial values are $S_i = 1_{6\times 6}$, $\upsilon_i = 0_{6\times 1}$ for $i = 1, 2$, and control parameters are selected as $\rho_i = 4I_6$, $\tilde{K} = 5I_6$. The sampling and communication frequency is 1000Hz.

As in [45], we also employ gain clipping as $\mu(t) = \min\{T/(T + t_0 - t) + 5, \xi\}$ with $\xi = 20$ for the real-robot experiment. This mitigates the instability issue caused by the measurement noise but introduces a small synchronization error. The experimental results are shown in Fig. 11-13 demonstrating that the position and velocity of the two manipulators achieve synchronization within $T + t_0$ while the control input remains bound.

Furthermore, we validate the robustness of the proposed PTS algorithm, as shown in Fig. 14-16, by setting different prescribed times T. These figures show that the finite settling time can be specified *a priori*. Additionally, the PTS algorithm demonstrates robustness to external disturbances by relying solely on time-varying gains, without requiring any compensatory techniques.

VI. Conclusion

In this paper, the PTS problem of networked Euler-Lagrange systems with external disturbances is addressed. We propose a novel prescribed-time PTS algorithm comprising distributed prescribed-time observers and local prescribed-time tracking controllers. A distinct feature of our algorithm is that the settling time of synchronization errors is independent of initial values and design parameters. Additionally, we analyze the properties of time-varying gains coupled with comparison functions to derive prescribed-time convergence. Our results require the leader to transmit its full state in the network. It would be interesting to consider the case where the leader can only transmit its partial state or output in the network, and thus a more general observer is needed.

Appendices

Proof of Proposition 2: Taking the time derivative of $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))$ yields $\mathrm{d}\big[(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))\big]/\mathrm{d}t = \Big(c_1(\gamma(\mu))^{c_1-1}\frac{\mathrm{d}\gamma(\mu)}{\mathrm{d}\mu}\dot{\mu} - c_2(\gamma(\mu))^{c_1+1}\Big)\kappa^{-c_2}(\gamma(\mu)) \leq (\varepsilon - 1)c_2(\gamma(\mu))^{c_1+1}\kappa^{-c_2}(\gamma(\mu)) < 0$, where we used (3) and (7). Therefore, $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))$ is a decreasing function of t for $t \in \mathcal{T}_p$. Note that $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))$ can be further expressed as

$$\begin{aligned}
&(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu)) \\
&\quad = (\gamma(a_0))^{c_1}\exp\left(\int_{t_0}^{t}\frac{c_1\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\Big|_{s=\mu(\zeta)}\dot{\mu}(\zeta) - c_2\gamma^2(\mu(\zeta))}{\gamma(\mu(\zeta))}\mathrm{d}\zeta\right) \\
&\quad \leq (\gamma(a_0))^{c_1}\exp\left(\int_{t_0}^{t}\frac{c_1 w_1\mu^{\iota_1}(\zeta)\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\Big|_{s=\mu(\zeta)} - c_2\gamma^2(\mu(\zeta))}{\gamma(\mu(\zeta))}\mathrm{d}\zeta\right) \\
&\quad \leq (\gamma(a_0))^{c_1}\kappa^{-(1-\varepsilon)c_2}(\gamma(\mu))
\end{aligned}$$

which means that $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))$ tends to zero as $t \to t_0 + T$. Therefore, $(\gamma(\mu))^{c_1}\kappa^{-c_2}(\gamma(\mu))$ is a $\mathcal{D}_T$ function. ■

Proof of Proposition 3: Taking the time derivative of $\dot{\gamma}(\mu)\kappa^{-c}(\gamma(\mu))$ yields

$$\begin{aligned}
&\mathrm{d}[\dot{\gamma}(\mu)\kappa^{-c}(\gamma(\mu))]/\mathrm{d}t \\
&\leq \dot{\mu}\kappa^{-c}(\gamma(\mu))\left(\frac{\mathrm{d}^2\gamma(\mu)}{\mathrm{d}\mu^2} + \frac{\mathrm{d}\gamma(\mu)}{\mathrm{d}\mu}w_2\dot{\mu}^{\iota_2-1} - c\frac{\mathrm{d}\gamma(\mu)}{\mathrm{d}\mu}\gamma(\mu)\right) \\
&\leq \dot{\mu}\kappa^{-c}(\gamma(\mu))\left(\frac{\mathrm{d}^2\gamma(\mu)}{\mathrm{d}\mu^2} + \frac{\mathrm{d}\gamma(\mu)}{\mathrm{d}\mu}w_2 w_1^{\iota_2-1}\mu^{\iota_1(\iota_2-1)}\right. \\
&\qquad \left. - c\frac{\mathrm{d}\gamma(\mu)}{\mathrm{d}\mu}\gamma(\mu)\right) \\
&\leq \dot{\mu}\kappa^{-c}(\gamma(\mu))(\varepsilon - 1)c\frac{\mathrm{d}\gamma(\mu)}{\mathrm{d}\mu}\gamma(\mu) < 0
\end{aligned} \tag{50}$$

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where we used (9). Therefore, $\dot{\gamma}(\mu)\kappa^{-c}(\gamma(\mu))$ is a decreasing function of t for $t \in \mathcal{T}_p$. Note that $\dot{\gamma}(\mu)\kappa^{-c}(\gamma(\mu))$ can be further expressed as

$$\begin{aligned}
&\dot{\gamma}(\mu)\kappa^{-c}(\gamma(\mu)) \\
&\le a' \exp\left(\int_{t_0}^{t} \frac{\frac{\mathrm{d}^2\gamma(s)}{\mathrm{d}s^2}\big|_{s=\mu(\zeta)}\dot{\mu}(\zeta)+\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}\ddot{\mu}(\zeta)}{\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}\dot{\mu}(\zeta)}\mathrm{d}\zeta\right) \\
&\quad \times \exp\left(\int_{t_0}^{t} \frac{-c\gamma(\mu(\zeta))\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}\dot{\mu}(\zeta))}{\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}\dot{\mu}(\zeta)}\mathrm{d}\zeta\right) \\
&\le a'\exp\Bigg[\int_{t_0}^{t}\dot{\mu}(\zeta)\Bigg(\frac{\mathrm{d}^2\gamma(s)}{\mathrm{d}s^2}\big|_{s=\mu(\zeta)}+\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}w_2(\dot{\mu}(\zeta))^{\iota_2-1} \\
&\quad - c\gamma(\mu(\zeta))\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}\Bigg)\Big/\left(\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(\zeta)}\dot{\mu}(\zeta)\right)\mathrm{d}\zeta\Bigg] \\
&\le a'\kappa^{-(1-\varepsilon)c}(\gamma(\mu)).
\end{aligned}$$

where $a' = \frac{\mathrm{d}\gamma(s)}{\mathrm{d}s}\big|_{s=\mu(t_0)}\dot{\mu}(t_0)$.

By Propositions 1 and 2, and $c_2/c_1 \le c/2$, we have $\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s} \le \varepsilon w_1^{-1}\frac{c}{2}\mu^{-\iota_1}\gamma^2(s)$ for $s \in [a_0, \infty)$, then $\gamma^2(\mu)\kappa^{-c}(\gamma(\mu))$ are $\mathcal{D}_T$ function and satisfies $\gamma^2(\mu)\kappa^{-c}(\gamma(\mu)) \le (\gamma(a_0))^2\kappa^{-(1-\varepsilon)c}(\gamma(\mu))$. We also note that $\frac{\mathrm{d}\gamma(s)}{\mathrm{d}s} \le \varepsilon w_1^{-1}c\mu^{-\iota_1}\gamma^2(s)$ for $s \in [a_0, \infty)$. Then $\gamma(\mu)\kappa^{-c}(\gamma(\mu))$ satisfies $\gamma(\mu)\kappa^{-c}(\gamma(\mu)) \le \gamma(a_0)\kappa^{-(1-\varepsilon)c}(\gamma(\mu))$. ■

Proof of Lemma 1: For $t \in \mathcal{T}_p$, applying comparison lemma (Lemma 3.4 in [39]) to (10) yields $V(t) \le \kappa^{-c_1}(\gamma_1(\mu))\exp\big(c_2(t-t_0)\big)V(t_0) + c_3\int_{t_0}^{t}\big[\exp\big(c_2(t-\zeta) - c_1\int_{\zeta}^{t}\gamma_1(\mu(s))\mathrm{d}s\big)\kappa^{-c_1}(\gamma_2(\mu(\zeta)))\big]\mathrm{d}\zeta$. The second term on the right-hand side of $V(t)$ satisfies

$$\begin{aligned}
&c_3\int_{t_0}^{t}[\exp\big(c_2(t-\zeta) - c_1\int_{\zeta}^{t}\gamma_1(\mu(s))\mathrm{d}s\big)\kappa^{-c_1}(\gamma_2(\mu))]\mathrm{d}\zeta \\
&\quad \le c_4\int_{t_0}^{t}\exp\big(-c_1\int_{\zeta}^{t}\gamma_1(\mu(s))\mathrm{d}s - c_1\int_{t_0}^{\zeta}\gamma_2(\mu(s))\mathrm{d}s\big)\mathrm{d}\zeta \\
&\quad \le c_4 T\kappa^{-c_1}(\gamma_1(\mu)) \qquad (51)
\end{aligned}$$

with $c_4 = |c_3|\exp\big(|c_2|T\big)$ where $\gamma_1(\mu) - \gamma_2(\mu) < 0$ is used. Using the inequality (51) yields $V(t) \le \kappa^{-c_1}(\gamma_1(\mu))\exp\big(c_2(t-t_0)\big)V(t_0) + c_4 T\kappa^{-c_1}(\gamma_1(\mu)) \le (c_5 V(t_0) + c_4 T)\kappa^{-c_1}(\gamma_1(\mu))$ where $c_5 = \exp\big(c_2 T\big)$. As a result, $\|x(t)\| \le (\underline{\lambda}_P^{-1}(\bar{\lambda}_P c_5\|x(t_0)\|^2 + c_4 T))^{\frac{1}{2}}\kappa^{-\frac{c_1}{2}}(\gamma_1(\mu))$. The proof is thus completed. ■

Proof of Lemma 2: Applying the comparison lemma to (11) obtains

$$\begin{aligned}
V(t) &\le \exp\big(-c_1\int_{t_0}^{t}\gamma(\mu(\zeta))\mathrm{d}\zeta\big)V(t_0) \\
&\quad + c_2\int_{t_0}^{t}\exp\big(-c_1\int_{\zeta}^{t}\gamma(\mu(s))\mathrm{d}s\big)\gamma(\mu(\zeta))\mathrm{d}\zeta. \qquad (52)
\end{aligned}$$

The second term on the right-hand side of (52) becomes

$$\begin{aligned}
&c_2\int_{t_0}^{t}\exp\big(-c_1\int_{\zeta}^{t}\gamma(\mu(s))\mathrm{d}s\big)\gamma(\mu(\zeta))\mathrm{d}\zeta \\
&= c_2\kappa^{-c_1}(\gamma(\mu))\int_{t_0}^{t}\exp\big(c_1\int_{t_0}^{\zeta}\gamma(\mu(s))\mathrm{d}s\big)\gamma(\mu(\zeta))\mathrm{d}\zeta \\
&= c_2\kappa^{-c_1}(\gamma(\mu))\int_{t_0}^{t}\exp\big(c_1\int_{t_0}^{\zeta}\gamma(\mu(s))\mathrm{d}s\big)\mathrm{d}\int_{t_0}^{\zeta}\gamma(\mu(s))\mathrm{d}s \\
&= c_2\kappa^{-c_1}(\gamma(\mu))c_1^{-1}\exp\big(c_1\int_{t_0}^{\zeta}\gamma(\mu(s))\mathrm{d}s\big)\big|_{t_0}^{t} \\
&= c_2 c_1^{-1}\kappa^{-c_1}(\gamma(\mu))(\kappa^{c_1}(\gamma(\mu)) - 1). \qquad (53)
\end{aligned}$$

Substituting (53) into (52) yields $V(t) \le \kappa^{-c_1}(\gamma(\mu))V(t_0) + c_2 c_1^{-1}(1 - \kappa^{-c_1}(\gamma(\mu)))$. As a result, $\|x(t)\| \le \sqrt{\underline{\lambda}_p^{-1}\bar{\lambda}_p\|x(0)\|^2 + \underline{\lambda}_p^{-1}|c_2|c_1^{-1}}$. Thus, (11) holds. ■

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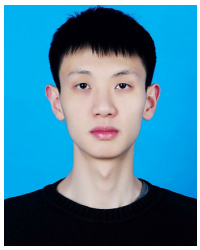

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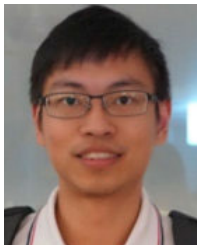

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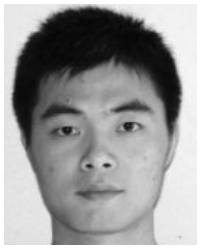

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